

Instructions Booklet

33rd World Puzzle Championship

Version : 1
Published on : 2026-09-26



THURSDAY, 15th OCTOBER

Session 1

01 Raghu Clan	09:00 - 10:00	60m	750 points	Individual
02 Vishwamitra	10:15 - 11:10	55m	650 points	Individual
03 Sita	11:25 - 12:00	35m	400 points	Individual

Session 2

04 Dasharatha's Dilemma	13:45 - 14:20	35m	400 points	Individual
05 Urmila	14:30 - 15:10	40m	450 points	Individual
06 Agastya	15:20 - 16:10	50m	600 points	Individual
07 Shoorpanakha	16:25 - 17:25	60m	1000 points	Individual
08 Team Round				Team

FRIDAY, 16th OCTOBER

Session 3

09 Janasthana	09:00 - 09:45	45m	600 points	Individual
10 Laxman Rekha	09:55 - 10:40	40m	400 points	Individual
11 Shabri	10:55 - 11:35	40m	450 points	Individual
12 Hanuman	11:45 - 12:20	35m	300 points	Individual

Session 4

13 Vali and Sugriva	14:00 - 14:50	50m	550 points	Individual
14 Squirrel	15:00 - 15:20	20m	200 points	Individual
15 Kumbhakarna	15:35 - 16:55	80m	1200 points	Individual
16 Team Round				Team
17 Team Round				Team

SATURDAY, 17th OCTOBER

Session 5

18 Indrajit	09:00 - 09:55	55m	650 points	Individual
19 Team Round				Team
20 Team Round				Team

Session 6

21 Team Round				Team
22 Ravana	15:15 - 16:15	60m	800 points	Playoffs



Competition Hall Rules

1. Each competitor has to sit at any of the pre-allocated desks of their respective teams in individual rounds. Teams have to work at their pre-allocated desks for team rounds.
2. Prior to the start of each round, competitors must ensure they are at their desks ready for the start of the round. Late arrivals may not be permitted to enter the competition hall to take part in a round at the discretion of the organizers.
3. Prior to the start of each round, competitors must clearly write their name, team and registration number on the front page of their competition booklet into the allocated space. If this information is not complete, then the organizers reserve the right not to award any points to that competitor for that round.
4. Competitors must not open their booklets before the official start of the round. When the signal for the start of the round has been given, competitors may open their booklets and begin solving the puzzles.
5. During each individual round, competitors have to remain silent, unless declaring completion of a round.
6. During team rounds, team members may talk to each other, unless it is stated otherwise in the round's note, but should do this with respect to other teams.
7. To declare a round complete, a competitor must close the competition booklet, clearly state 'finished' and raise an arm with the booklet. The competitor's arm must be raised until the booklet is collected. The same rules apply for the team rounds.
8. Competitors or teams who complete a round more than five minutes in advance, are allowed to leave the competition hall quietly. Competitors or teams who complete a round with five minutes or less left are not allowed to leave their desks to not cause unnecessary disruption to fellow competitors.
9. Competitors who leave the competition hall for any reason will be not allowed to continue in that round.
10. When the signal to finish a round is given, competitors have to stop solving immediately, close their booklets, put their pens/pencils down and their hands up with their booklets for collecting.
11. At the end of a round, competitors have to remain seated until all booklets have been collected.
12. Mobile phones and electronic devices are not permitted to be used in the competition hall during the rounds. The devices have to be turned off and must not be placed on the competitor's desk.
13. Only team captains and official observers equipped with a name tag are allowed to enter the competition hall while either individual or team rounds are taking place. Other non-competing participants may enter the competition hall at the discretion of the organizers.
14. Competitors may not use cameras or other recording devices during rounds. Only observers may do so, at the discretion of the organizers. They have to respect the competitors and not use flash photography or cameras with excessive sounds.
15. When a competitor believes that there is a problem with a puzzle, they must clearly state that puzzle is wrong by writing 'Wrong puzzle' next to it. The competitor must not notify the organizers during the round. This will be investigated upon completion of the round.
16. Puzzles can be completed in any order within a round, unless it is stated otherwise in the round's note. The points' value of a puzzle is an indication of its expected difficulty, although individual solving experience may differ. The difficulty of an example puzzle does not necessarily reflect the difficulty of the corresponding competition puzzle.
17. The boxes above each puzzle are reserved for markers' notes. Competitors must not write in the boxes.
18. Permitted items which can be used in the competition hall, unless stated otherwise, are: pens, pencils, pencil sharpeners, erasers, rulers, scales, blank papers and instruction booklets annotated with notes regarding puzzle instructions and preparation notes.
19. Drinks and snacks are permitted as long as they do not disturb other competitors with a strong smell or rustling packet.
20. It is strictly forbidden to use electronic devices such as music players or headphones or any type of calculator. Use of such equipment may lead to the disqualification of the competitor.
21. Any other items brought into the hall must be kept in a bag on the floor and placed under the competitor's desk, so as not to block the aisles.

22. When a round has been evaluated, fully marked booklets are returned to a team member of the respective country.
23. In case of any query after a booklet has been returned to a competitor, the query must be raised within the specified time. The booklet should be left with the organizers for investigation.
24. Puzzles may be photographed during the marking phase in order to prevent subsequent interventions.
25. Team captains are responsible for ensuring that any information given to them related to the competition is effectively relayed to their team.
26. In case of a major mistake in one of the rounds, organisers reserve the right to cancel the round, either by removing it from the schedule, or by not rewarding any points for it to any of the competitors.
27. The official competition booklets will not contain examples given in the instruction booklet. Therefore, we recommend to bring the Instruction Booklet, which contains an example of every puzzles that will be part of the championship.
28. In the team rounds, the official competition booklets may not contain the instructions of puzzles, only the names. It is advised to bring at least one Instruction Booklet for a team for these rounds.
29. In any case of inconsistency between the instruction booklet and the official competition booklets, e.g. instructions or points, the information in the final version of the instruction booklet will be considered valid.
30. In the competition hall, a timer counting down to the end of the round will be visible for all the competitors.

Scoring and Bonuses

1. Points will be awarded only for fully and correctly solved puzzles. In general, there is no partial points unless it is stated otherwise in the round's note.
2. In individual rounds, the bonus points for a round for each full remaining minute will be awarded to any competitor who correctly solves all the puzzles in the round.
3. In team rounds, the bonus points for a round for each full remaining minute will be awarded to any team who correctly solves all the puzzles in the round.
4. At the judge's discretion, 0.8x bonus, rounded to the closest integer, will be awarded in the case of a single minor mistake in no more than one Puzzle. A minor mistake is considered as at most two incorrectly represented cells in at most one of the Puzzles. Any variance from this is noted in individual round notes.

Any breach of these rules may lead to a competitor or team being disqualified from the round or competition.

The decision of the tournament directors is final.

For puzzles with partial points, the points will be awarded only if it is part of the overall solution.

Acknowledgements

All the examples in the instructions booklet were made by the organizing team with certain examples sourced from:

1. Past WPC Instruction booklets
2. Past LMI Contests
3. Puzzle Innovation contest
4. Puzzle blogs of organizing team

They cannot be commercially used. All rights have been reserved.

Most images in the championship are drawn using Penpa+. We would like to thank:

1. The original creator opt-pan for penpa edit - <https://opt-pan.github.io/penpa-edit/>
2. Swaroop Guggilam for his recent efforts in adding features to Penpa-edit - <https://swaroopg92.github.io/penpa-edit/>

Parts of the Competition Rules above are reused from Instructions Booklets of previous WPCs.

Glossary

Odd and Even

Odd digits are 1, 3, 5, 7, 9.

Even digits are 2, 4, 6, 8.

Parity

Even and odd are different parities. Two digits have the same parity if they are both even or both odd.

Prime

Prime digits are 2, 3, 5, 7

Adjacent and Neighbouring

Cells sharing an edge are adjacent. A cell can have a maximum of four adjacent cells.

Cells sharing an edge or a corner are neighbouring. A cell can have a maximum of eight neighbouring cells.

MxN region/grid

A MxN region/grid is a region/grid with 'M' rows and 'N' columns.

Touching

Cells that share an edge or a corner touch each other, and this applies to any contents placed in those cells.

Queens

A Queen, as in chess, can move in a straight line in any of the eight directions, Top, Top-Right, Right, Bottom-Right, Bottom, Bottom Left, Left and Top-Left, in a straight line of cells, starting from the first cell touching the queen cell in that direction.

Queens can move for any length unless certain things mentioned in the rules block the movement.

Single connected area

This phrase is used repeatedly in puzzles with either shading or white cells that must adhere to the phrase. The phrase means that from any cell in the group described, it must be possible to visit any other cell in the group just by traveling along grid edges through other cells in the group.

Loops

In most loops, the following line is used – 'Draw a closed loop passing through the centres of cells horizontally and vertically'. In all such cases the loop passes through grid-edges via the centres of cells. Unless explicitly mentioned, the following rules are inclusive of a loop that does not cross, overlap or intersect itself. In general, if these behaviours are not mentioned, they are not allowed. A slight variation to the above has 'Draw a closed loop passing through the centres of all cells horizontally and vertically', and this means all cells the loop can visit within the rules provided, must be visited. This is not necessarily the case when the word 'all' is not given, but it can happen then too as part of solving.

Consecutive

Two digits are consecutive if their difference is 1. If the word is used in the context of cells, it is a series of cells in the same row or column sharing grid edges at all points across the series.

2x2 group/area of cells

This phrase is used to describe the four cells immediately touching any given point in a typical square grid. If there are substitutions for the same in irregular grids, they are explicitly explained in individual puzzles.

Checkerboard

A checkerboard pattern is a 2x2 area of cells where the top-left and bottom-right cells are of one type and the top-right and bottom-left cells are of another type.

Dividing the grid into regions

Unless explicitly mentioned, wherever this kind of phrasing comes up, the regions must contain full cells within them, and not partial cells. Also, there will not be a requirement for any unnecessary bold lines that do not connect to close up a region.

Rotation and reflection are allowed

This phrase is usually used in puzzles where a given list of shapes or objects including words and numbers are required to be placed into the grid. It means that even the rotated version or the reflected version of the given shape is considered to be the shape itself replicated in the grid, so it satisfies the instance of that shape appearing in the bank.

Across or down

This phrase is used generally in word puzzles and it means that a word must only be placed in such a way that it can be read, as it is read in the given list, from left to right or top to bottom. For instance, a word 'Puzzle' cannot be placed in the grid as 'elzzuP' if this phrase is used.

Corresponding row or column

This phrase is generally used in puzzles with outside clues. For a clue outside the left of the grid, the row must be used to satisfy it, and if order/direction is specified then it has to be satisfied from left to right, if not, it can be satisfied from right to left too. Similarly, for a clue outside the top (or bottom) of the grid, the column must be used to satisfy it.

Visibility/looking/etc.

A lot of puzzles have a characteristic of 'visibility', and the rules may say that certain conditions must be met when 'looking from' a point, or some other variation of this phrase. In all cases, it means that if an observer was viewing the cells in that direction in a straight line (row and/or column), the observer's vision would capture the given conditions. In some cases where counting is involved, this comes with the phrase 'including the cell itself', and in such cases the observer is placed in a cell and can observe in horizontal and vertical directions till the specified vision blockers, and can of course observe their own cell too, so all such cells get added to the count for the observer.

01 Raghu Clan

750 points

60 Minutes

10x Bonus

Individual

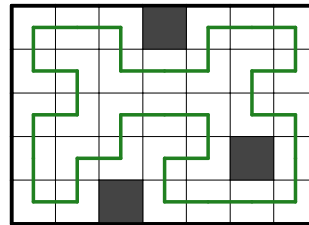
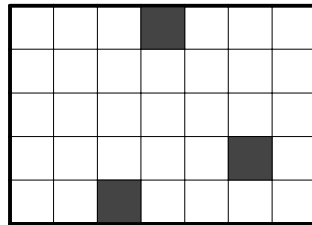
Raghu Clan delves into the lineage and ancestry of Rama, the prince of Ayodhya and his brothers, who serve as the central figures of Ramayana.

Mirroring this theme of origins, this round serves as an introduction to the inaugural few World Puzzle championships which had their origins in crosswords and language-based puzzles. Additionally, this segment features a collection of logic puzzles, which are now fondly referred to as casual puzzles and were prominent in the early years.

1. Simple Loop

15 points

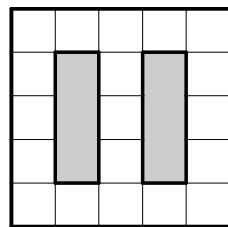
Draw a single non-intersecting loop of horizontal and vertical segments that passes through all of the white cells exactly once.



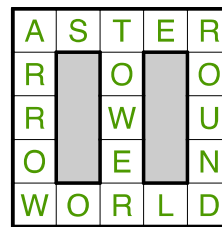
2. Criss Cross

45 points

Place all the listed words in the given grid going across or down.



ARROW
ASTER
ROUND
TOWER
WORLD



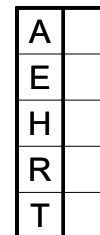
3. Letter Weights

90 points

Write a number next to each letter (in each cell) so that the numbers corresponding to the letters in each given word have the given sum. Different letters must have different numbers. The range of numbers is given.

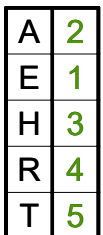
{1~5}

RAT 11
TEAR 12
ATE 8
THE 9
HAT 10



{1~5}

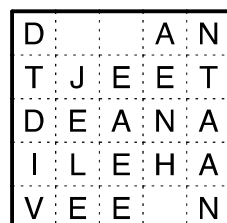
RAT 11
TEAR 12
ATE 8
THE 9
HAT 10



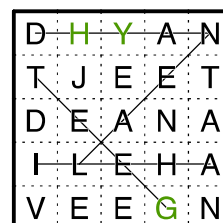
4. Word Search

80 points

Find all the words from the list in the grid of letters. Each word reads in a straight line in any direction: horizontally, vertically or diagonally. Some cells have been left blank and can be filled with exactly one letter each to help find all the words.



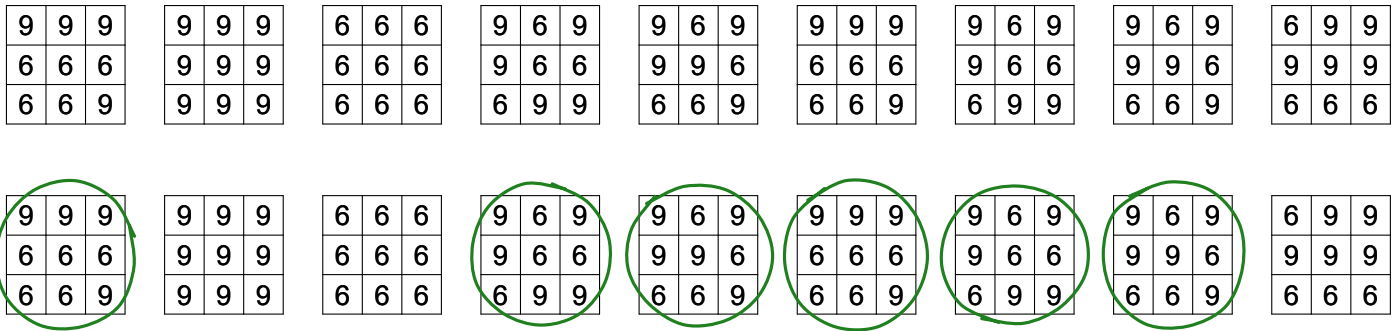
AHELI
DHYAN
GEET
NEAL



5. Find Pairs

30 points

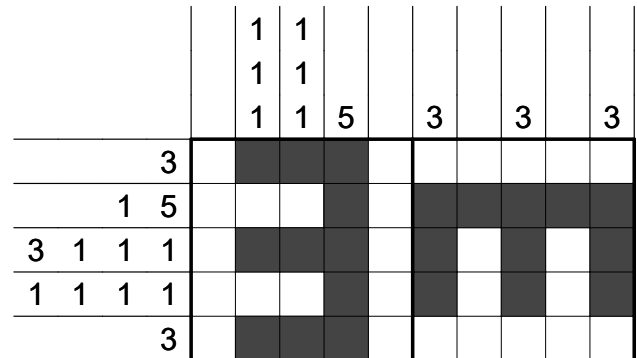
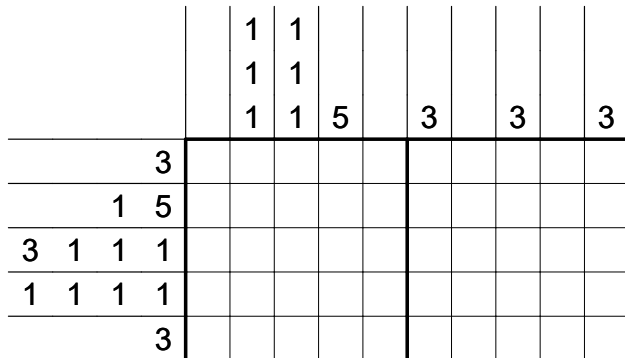
Find and mark three identical pairs among the given images. Rotated / reflected images are not considered as identical.



6. Nonogram

45 points

Shade some cells so that the numbers outside the grid indicate the groups of consecutive black cells which are in that row/column in order, either from left to right or from top to bottom. There must be at least one white cell between any consecutive shaded groups. Rows and columns without outside clues can have any pattern of shaded and unshaded cells.



7. Diagramless Crossword

200 points

Place the given set of words in the grid such that all words must be connected to each other and no additional words (even of two letters) can be formed except for the given words. The given Across and Down clues indicate the length of the word. Clues are numbered sequentially in the grid in reading order from left to right and top to bottom. The final diagram formed must have rotational symmetry. The numbers inside the grid indicate the positions of those across / down clues.

ACROSS

1 4

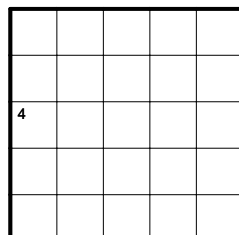
4 5

5 4

DOWN

2 5

3 5



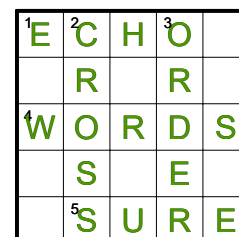
ECHO

SURE

CROSS

ORDER

WORDS



8. Chronological Order

60 points

Arrange the images in a chronological order. Enter the letter next to the image in the corresponding box given below the puzzle.



A



B



C



D

C	A	D	B
1	2	3	4

9. Counting

15 points

Count the number of rectangles (including squares) in the figure.



5

10. Long Division

105 points

Identify which letter stands for which digit such that it forms a correct long division computation. Each letter stands for one digit (0–9) and no two different letters can represent the same digit. The first digit of any multi-digit number, divisor, quotient, dividend, or intermediate step cannot be zero.

$$\begin{array}{r}
 \text{O N E} \\
 \text{S I X} \overline{) \text{E I G H T}} \\
 \underline{- \text{S I X}} \\
 \text{I X G H} \\
 \underline{- \text{T R T X}} \\
 \text{R S H T} \\
 \underline{- \text{R S H X}} \\
 \text{T}
 \end{array}$$

$$\begin{array}{r}
 \text{O N E} \\
 \text{S I X} \overline{) \text{E I G H T}} \\
 \underline{- \text{S I X}} \\
 \text{I X G H} \\
 \underline{- \text{T R T X}} \\
 \text{R S H T} \\
 \underline{- \text{R S H X}} \\
 \text{T}
 \end{array}$$

E	G	H	I	N	O	R	S	T	X

E	G	H	I	N	O	R	S	T	X
9	8	7	3	4	1	5	6	2	0

11. Mastermind

40 points

Find out the correct series of letters with the help of the information given by the black and white dots. Black dots imply correct letters in the correct position while white dots imply correct letters in the wrong position. Letters are not repeated in the clues or in the solution.

A E I O U ◯ ◯
 L Y R I C ◯
 V O C A L • • •
 V O I C E • • ◯
 L O W E R • • • ◯

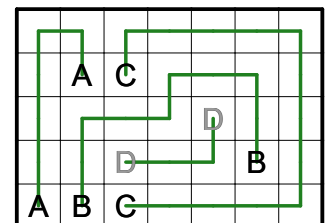
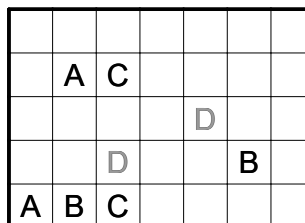
A E I O U ◯ ◯
 L Y R I C ◯
 V O C A L • • •
 V O I C E • • ◯
 L O W E R • • • ◯

V	O	W	E	L

12. Arukone

25 points

Connect every pair of letters through horizontal and vertical paths so that the paths cover all cells and no two paths cross or overlap each other.



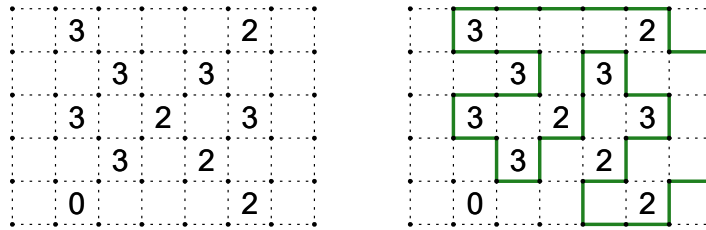
Prince Rama and his three brothers were raised under the tutelage of Sage Vasishta, the counselor to their father King Dasharatha. Later, Vishwamitra mentored Rama, transitioning him from a sheltered prince into a formidable warrior. He taught him archery and military strategy, instilling vital principles of righteousness.

Comprising 18 puzzles across a range of easy, medium, and hard difficulty, this round showcases 6 distinct classic puzzle categories that have remained an integral part of WPCs over the years.

1, 2, 3. Slitherlink

20 + 25 + 65 points

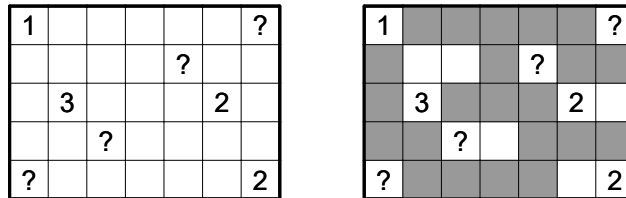
Connect some pairs of orthogonally adjacent dots to form a single non-intersecting loop. Clues represent the number of edges drawn surrounding the clue (up to four).



4, 5, 6. Nurikabe

15 + 20 + 40 points

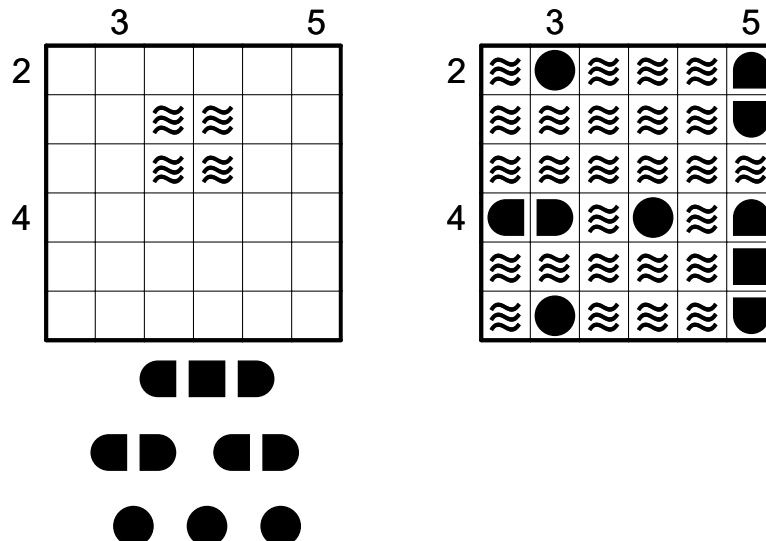
Shade some cells so that all shaded cells form one orthogonally connected area and no 2x2 region is entirely shaded. Clues cannot be shaded, and every orthogonally connected area of unshaded cells contains exactly one clue, the value of which represents the size of the area. A question mark "?" as a clue can represent any positive number.



7, 8, 9. Battleships

15 + 35 + 45 points

Place the given fleet of ships with the shapes of the ships as shown. The numbers outside the grid indicate the number of cells occupied by a ship segment in that row or column. Ships cannot touch each other, even diagonally. Some cells are known to be water unoccupied by ships, and are indicated by waves.



15 + 35 + 65 points

10				
8				
6				
4	8			3
				10
				4

10	10	10	10	10	1
8	8	8	8	10	3
6	6	8	8	10	3
4	6	8	8	10	3
4	6	6	4	10	10
4	4	6	4	4	4

10 + 30 + 30 points

(A, B, C)

A 4x4 grid with labels A, B, and C on the top and bottom edges, and A and B on the left and right edges.

	A	B	C	
A	A	C	B	
			A	B
	B			C
	C	B		A
A		A	C	
				B
	A	A		

30 + 75 + 80 points

		17	13	26	27
	10				
23					
8					
17					
23					

		17	13	26	27
	10	1	4	2	3
23	1	2	9	3	8
8	3	5		8	9
17	2	3	1	4	7
23	5	6	3	9	

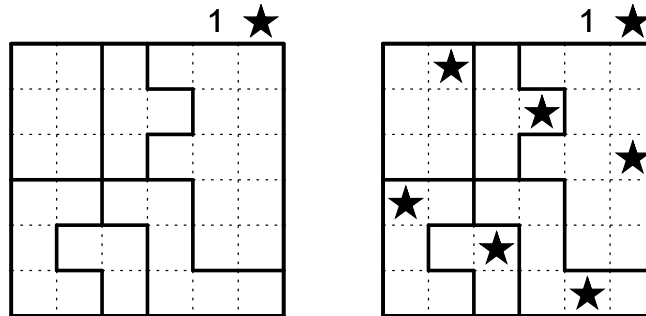
Upon finishing his education, Rama was guided by Vishwamitra to Mithila, ruled by King Janaka, to participate in his daughter Sita's Swayamvara- a ceremony where the bride chooses her spouse from assembled suitors. Rama successfully performed the onerous task of stringing Lord Shiva's bow, thereby earning Sita's hand in marriage. In a grand ceremony, Rama's three brothers were also wed to Sita's three sisters.

To honor the marriage of these four couples, this round presents four pairs of puzzles spanning different categories, alongside four hybrid puzzles combining the puzzles in each pair.

1. Star Battle

15 points

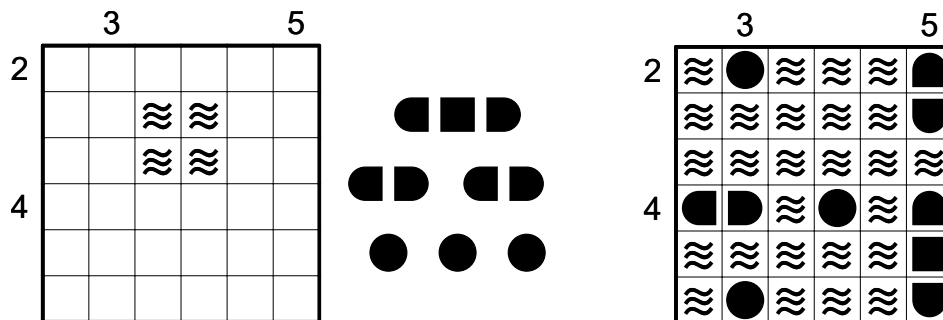
Place stars into some cells such that each row, column, and outlined region contains given number of stars. Stars may not touch one another, not even diagonally.



2. Battleships

25 points

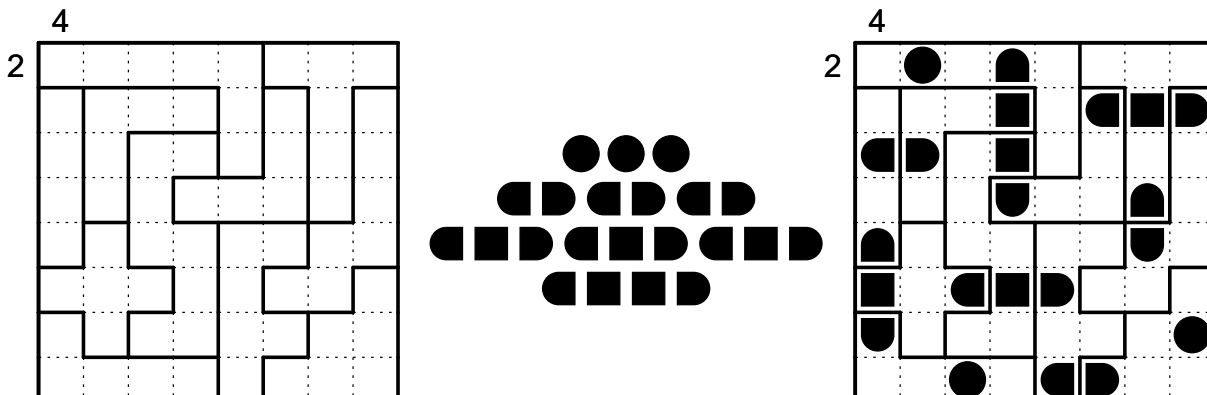
Place the given fleet of ships with the shapes of the ships as shown. The numbers outside the grid indicate the number of cells occupied by a ship segment in that row or column. Ships cannot touch each other, even diagonally. Some cells are known to be water unoccupied by ships, and are indicated by waves.



3. Sea Battle

30 points

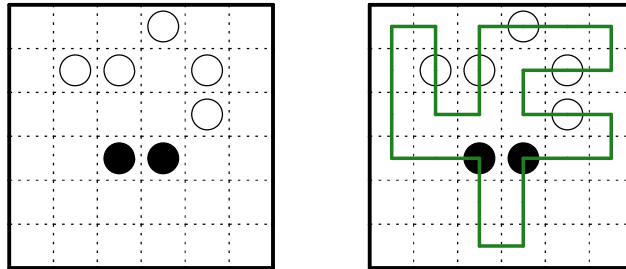
Place the given fleet in the grid so that every row, column and boldly marked area contain parts of exactly two different ships. Ships don't touch each other, not even diagonally. Ships may cross over bold lines. A clue outside the grid indicates the number of cells in the corresponding row or column that are occupied by ship segments.



4. Masyu

25 points

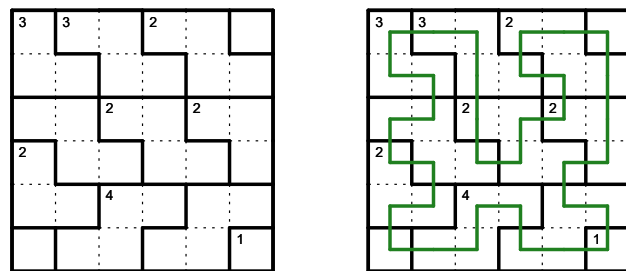
Draw a non-intersecting loop through the centers of some cells that passes through every circle. The loop must turn on black circles and travel straight through the cells on either side. The loop must go straight through white circles, and turn in at least one of the cells on either side.



5. Maxi Loop

40 points

Draw a non-intersecting loop through the centers of all cells. A number in a region represents the number of cells occupied by the largest continuous loop segment within the region.

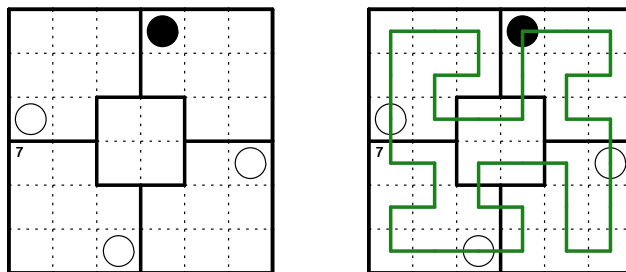


6. Maxi Masyu

30 points

Draw a non-intersecting loop through the centers of all cells that passes through every circle. The loop must turn on black circles and travel straight through the cells on either side. The loop must go straight through white circles, and turn in at least one of the cells on either side.

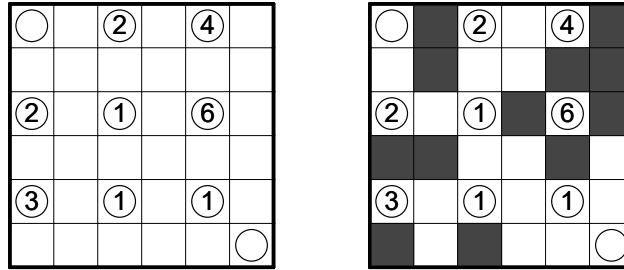
A number in a region represents the number of cells occupied by the largest continuous loop segment within the region.



7. Kurotto

65 points

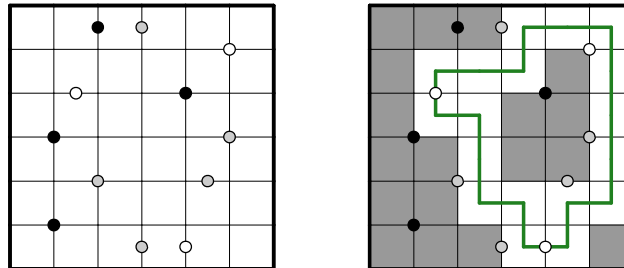
Shade some cells so that clues represent the total size of the orthogonally connected areas of shaded cells that share an edge with the clue. Clued cells cannot be shaded.



8. Kurarin

25 points

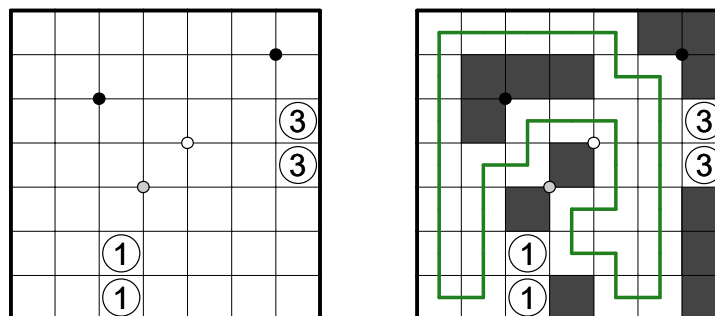
Shade some cells and draw a non-intersecting loop through the centers of all remaining cells. A black dot must touch more shaded cells than unshaded ones. A white dot must touch more unshaded cells than shaded ones. A grey dot must touch an equal amount of each.



9. Kurarin Kurotto

35 points

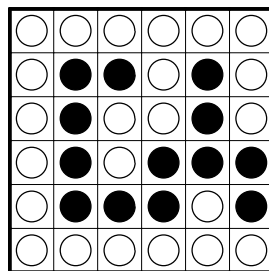
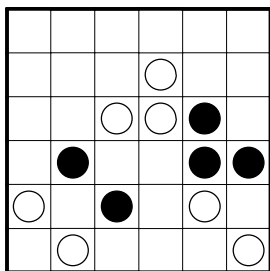
Shade some cells and draw a non-intersecting loop through the centers of all remaining white cells without a number. A black dot must touch more shaded cells than unshaded ones. A white dot must touch more unshaded cells than shaded ones. A grey dot must touch an equal amount of each. Number clues represent the total size of the orthogonally connected areas of shaded cells that share an edge with the clue (there can be shaded groups other than these too). Numbered cells cannot be shaded.



10. Yin-Yang

25 points

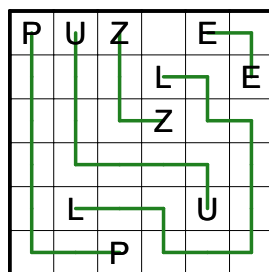
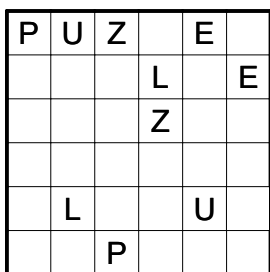
Place a circle into each cell of the grid - some white and some black - such that all circles of the same type must lie in cells forming one orthogonally connected area. No 2x2 region may contain all one type of circle.



11. Alphabet Link

20 points

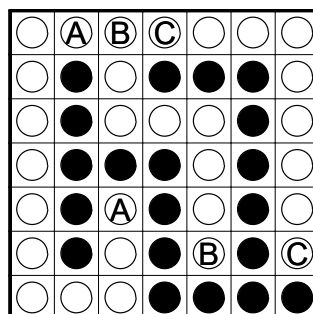
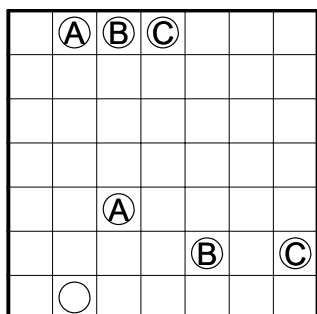
Draw non-intersecting paths through the centers of some cells, each connecting one clue to its equal counterpart.



12. Yin-Yang Link

65 points

Place a circle into each cell of the grid - some white and some black - such that all circles of the same type must lie in cells forming one orthogonally connected area. No 2x2 region may contain all one type of circle. Draw paths through the centre of all the white circles, each connecting one clue to its equal counterpart.



King Dasharatha intended to coronate the virtuous Rama as the king of Ayodhya. However, Kaikeyi, his third wife, invoked two boons granted to her in the past - to banish Rama and crown her own son Bharata instead. Deeply attached to both sons, Dasharatha faced an agonizing conflict of whether to choose Rama out of his paternal affection or uphold his promise by selecting Bharata. This round mirrors that profound choice, presenting you with a dilemma: will you tackle slightly complex puzzles governed by simple rules, or opt for somewhat simpler puzzles bound by complex rules?

Round Rules

In this round, there are 7 pairs of puzzles from different puzzle categories. In each pair participants will get credit for only one of the two puzzles. In each pair, both the puzzles have been assigned equal points.

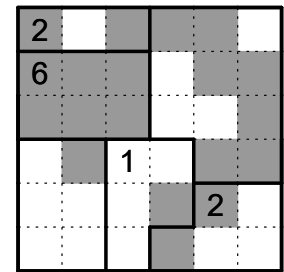
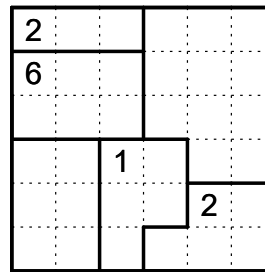
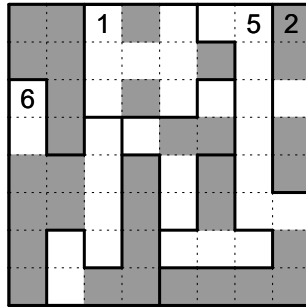
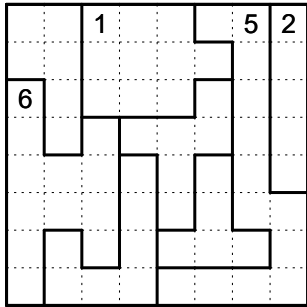
1. Clone Wars

70 points

1. Aqre

Shade some cells, such that there is no 2x2 area of unshaded cells. Numbers give the size of shaded group/s in the region. There are two possibilities for each region. If a region has only one shaded group, it orthogonally touches with exactly one other group of the same shape from a different region. Or if there are multiple groups of the same shape within the region that do not orthogonally touch shaded groups in any other region. Rotations and reflections count as same shape.

OR Shade some cells so that all shaded cells form one orthogonally connected area. Regions with numbers must contain the indicated number of shaded cells. There may not exist a run of more than three consecutive shaded or unshaded cells horizontally or vertically anywhere in the grid.



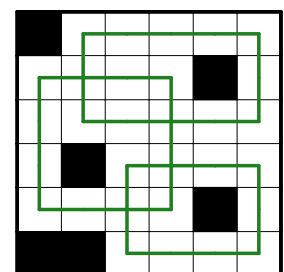
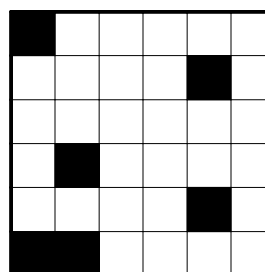
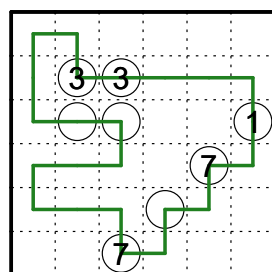
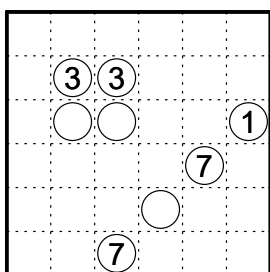
2. Bhai Bahan

50 points

2. Ringring

Draw a non-intersecting loop through the centers of some cells that passes through every circle. Consider the loop split into segments, such that within a segment, it either turns in each cell or goes straight in each cell, and segment types alternate along the path. When two circles are orthogonally adjacent, they belong to different types of segments (even if they are not directly connected along the loop). Numbered circles indicate the length of the current segment.

OR Draw rectangular loops through the centers of empty cells so that every empty cell gets used. The sides of different rectangles may intersect each other, but not turn at their intersection or otherwise overlap.



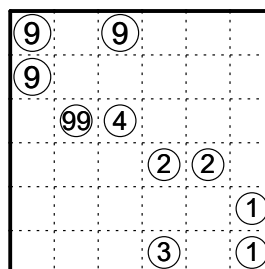
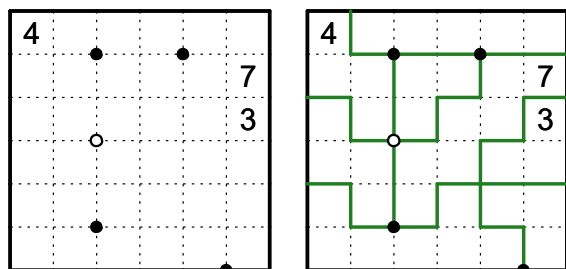
3. Heavy Dots

70 points

3. Araf

Draw vertical/horizontal lines between dots to divide the grid into regions. A region may contain at most one number, equal to the area of the region. No region may cover a 2×2 area. Some dots are “heavy” and have 3 or 4 lines connected to them. Black circles indicate heavy dots with exactly three lines; white circles indicate heavy dots with four lines. Not all heavy dots are given, but there can be no heavy dots orthogonally adjacent to the indicated ones.

OR Divide the grid into regions of orthogonally connected cells. Each region must contain exactly two circles and have an area that lies strictly between the two numbers in the circles.



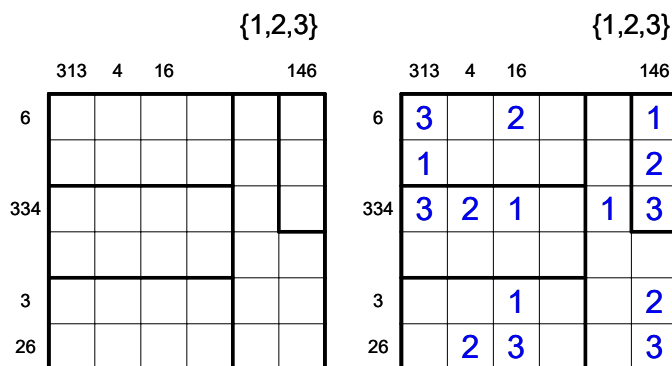
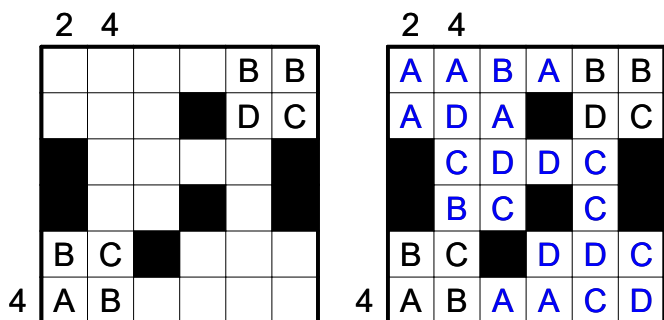
4. Candy Crush

60 points

4. Summon

Place a letter from A-D into each white cell such that there exist no horizontal or vertical runs of three or more consecutive cells containing the same letter. It must be possible to make any cell with a letter become part of such a run by making only one swap of two orthogonally adjacent letters. Numbers outside the grid indicate the number of distinct alphabets in the respective row/column.

OR Place numbers into some cells such that cells with the same digit don't touch, not even diagonally, and each region contains exactly the range of digits outside the grid. Groups of consecutive cells containing digits in a row or column form numbers by reading the digits left to right or top to bottom, respectively. Clues outside the grid indicate the sum of such numbers in the corresponding row or column.

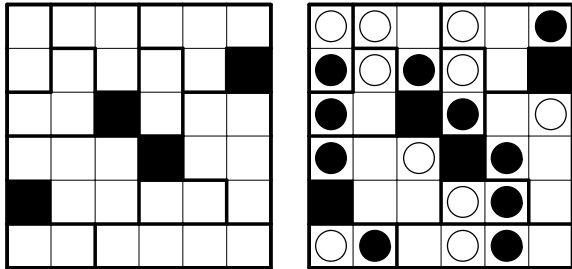


5. Dosun Fuwari

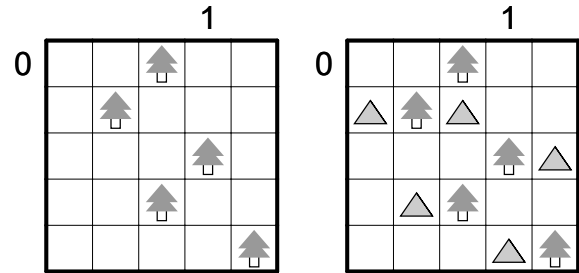
30 points

5. Tents

Place one iron ball (black circle) and one balloon (white circle) in each region. Iron balls are heavy and sink, so they must only be placed directly above the bottom edge of the grid, a black cell, or another iron ball. Balloons are light and float, so they must only be placed directly below the top edge of the grid, a black cell, or another balloon.



OR For each tree in the grid, place a tent in an empty orthogonally adjacent cell, connecting to it. Tents may not touch one another, not even diagonally. A clue given outside the grid represents the number of tents in the corresponding row or column.



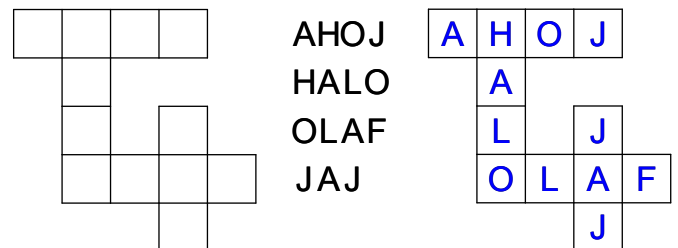
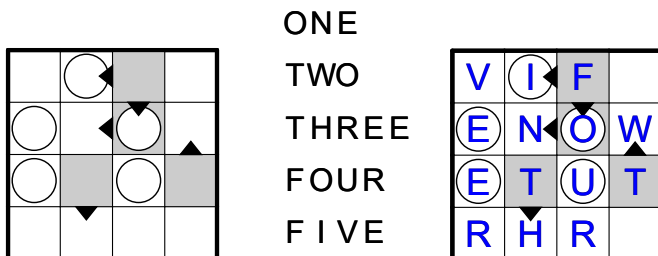
6. Tangled Words

70 points

6. Criss Cross

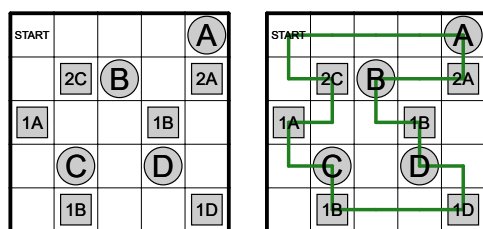
Place the given words into the grid, according to the following rules: For any word each letter must be placed horizontally or vertically adjacent to the preceding letter. Each word starts in a grey cell and the next letter must be placed in the direction of the arrow. Words may use the same cell more than once; however, if a word contains a double letter (the same letter twice in a row), it has to use two different cells for these. Multiple words can start in the same grey cell. A letter in a grey cell must be the starting letter for at least one word. Cells with circles contain vowels (A, E, I, O, U), and vowels must be placed only in cells with circles. Cells may remain empty. Each letter must be used by at least one word.

OR Place all the listed words in the given grid going across or down.



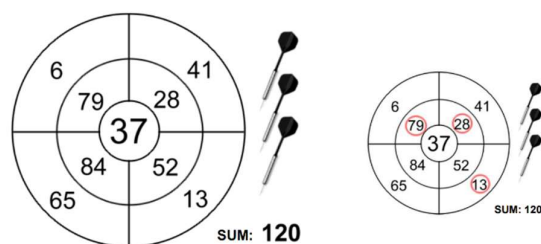
Find the path of the transporter so he can deliver all the transport orders. The path must begin in the cell marked START, run horizontally and vertically and finish in the starting cell again, without using any cell more than once. The squares represent objects that have to be delivered to their respective destinations, namely the grey circles. The letters indicate where the object have to be delivered. Whenever the transporter reaches a cell with a square, he picks up an object; Whenever he reaches a destination, he unload all the objects with the matching letter. On his path, the transporter has to pick up all the objects and deliver as well.

The numbers inside the square indicates the size of the object. The transporter has a limited capacity that cannot be exceeded at any time. The capacity of the transporter is given below the grid. It is sufficient to draw the path without the direction.



Capacity = 3

OR Use all the darts / arrows to score the given target exactly. No section of the board may be hit more than once.



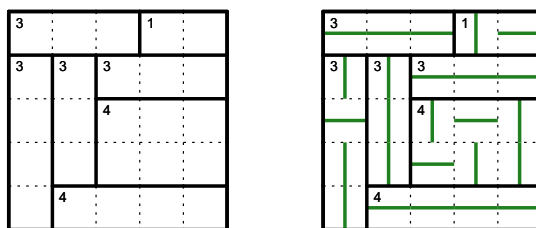
When Laxman left for the forest with Rama and Sita, his wife stayed back at his request, prioritizing her husband's duty. She traded her wakefulness with Laxman, sleeping continuously for 14 years of exile to allow him to stay awake and serve his brother Rama day and night without distraction.

In the spirit of Urmila's silent sacrifice, this round honors puzzles that the authors regard as unsung heroes—elegant designs that failed to receive the attention they truly deserved.

1. Juosan

90 points

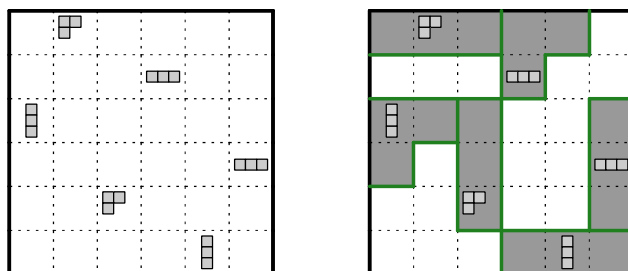
Place a horizontal or vertical line segment into each cell, connecting the centers of two opposite edges of the cell. A number in a region represents how many horizontal or vertical line segments it contains - whichever there's more of. There may not exist a run of three or more consecutive cells containing distinct parallel line segments anywhere in the grid.



2. It's Yours

35 points

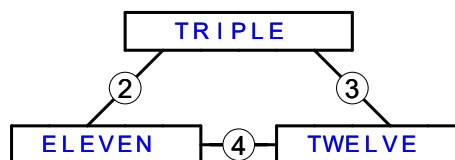
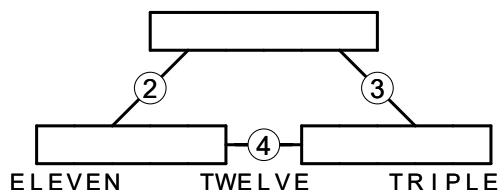
Place polyominoes into the grid such that each polyomino contains exactly one clue and touches exactly one other polyomino orthogonally. The clue inside a polyomino shows the exact shape of the polyomino it touches, without rotations or reflections.



3. Common Letters

60 points

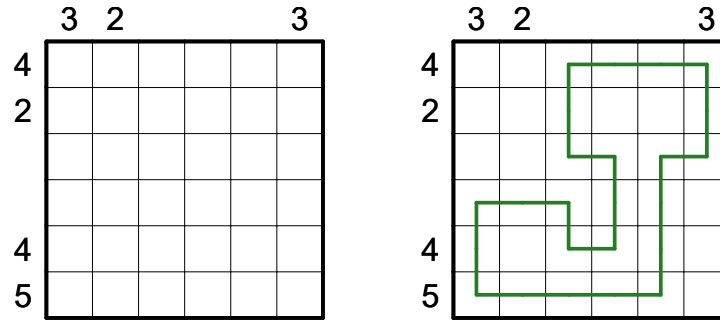
Place each word into a different box such that the number between two boxes indicates the number of letters those words have in common. Duplicate letters are counted separately.



4. Catloop

40 points

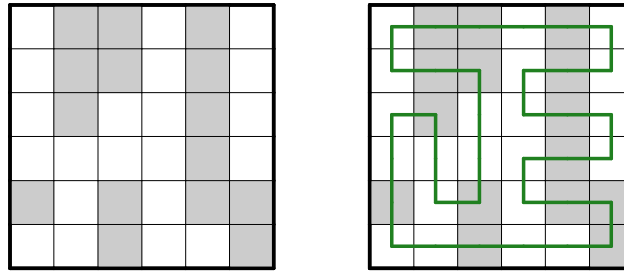
Draw a continuous loop consisting of straight sections into the grid in a way that the loop must not touch and cross itself. The numbers outside the grid show how many squares are used in that row or column by the loop.



5. Land and Sea

15 points

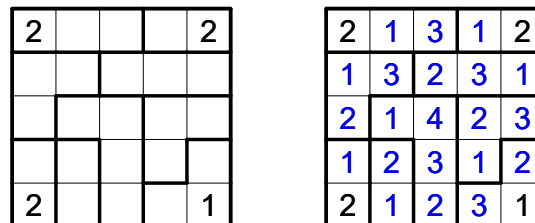
Draw a closed loop through all cells in the grid horizontally and vertically. The loop can't run through more than 2 white cells consecutively.



6. Largest Number

55 points

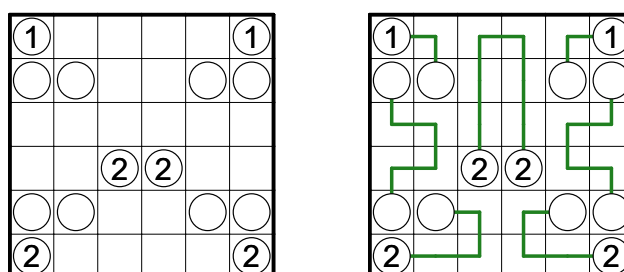
Place a number into each cell so that each region contains the numbers from 1 to N with no repeats, where N is the number of cells in the region. Numbers of the same value may not touch one another orthogonally. Two adjacent numbers may not both be the largest number of their respective regions.



7. Mintonette

40 points

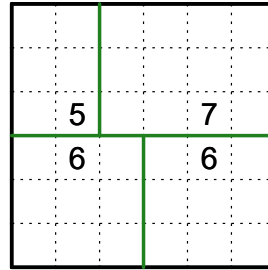
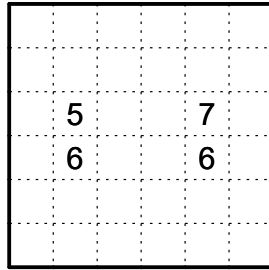
Connect circles in pairs by drawing a line that goes horizontally and vertically through the centers of cells. Lines cannot touch or cross themselves or each other. If a circle contains a number, it represents the number of turns for the line between two circles. If two circles have no number, a line may make any number of turns between circles. All cells must be used by the lines, and each circle must be connected to another circle.



8. Recto

65 points

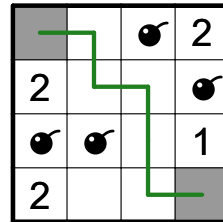
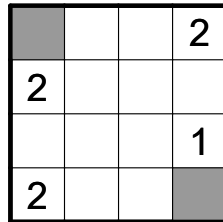
Divide the grid into rectangular regions of orthogonally connected cells. Each region must contain exactly one clue, which indicates the sum of the width and height of its rectangle.



9. Mine Field

30 points

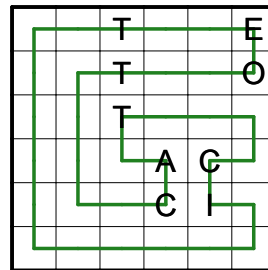
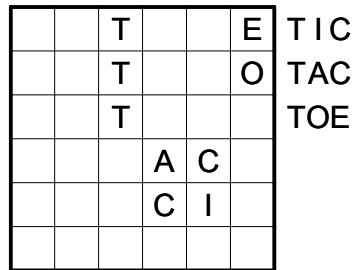
Draw a path between the two shaded cells connecting squares orthogonally. Numbers show how many mines are present in the surrounding 8 cells. The path cannot cross over cells containing mines or numbers, and it does not need to visit all cells without mines.



10. Sutuna Loop

20 points

Draw a closed loop passing through the centres of all cells horizontally and vertically. All words from the given list must be read along the loop only in a single direction in the given order.



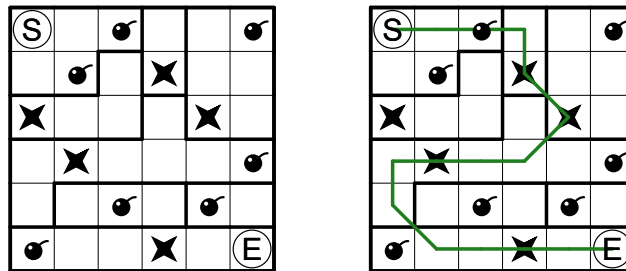
During his travels in the forests Rama stayed in the hermitage of Sage Agastya. Agastya bestowed Rama with several divine weapons like the sword and bow of Lord Vishnu, inexhaustible quivers, an impenetrable armor and Lord Brahma's mighty weapon.

This round features innovative puzzle types that represent the celestial weapons acquired by Rama.

1, 2. Nagastra

20 + 55 points

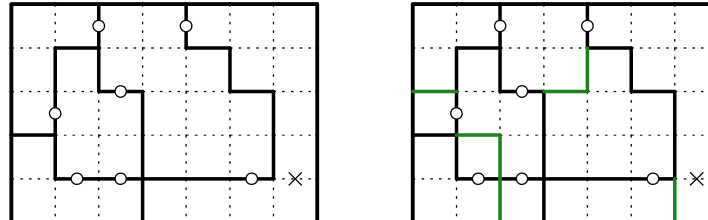
Draw a non-overlapping and non-intersecting path from cell 'S' to cell 'E' passing through cell centers without forming acute angles or touching itself orthogonally or diagonally. The path passes through weapons (mines and stars) along the way so that a weapon must be used exactly once before passing through another one. The path can only pass through the edge of a region using a mine. The path can only pass through a vertex of any cell (irrespective of regions) using a star. The path may end with an unused weapon.



3, 4. Box and Blob

25 + 55 points

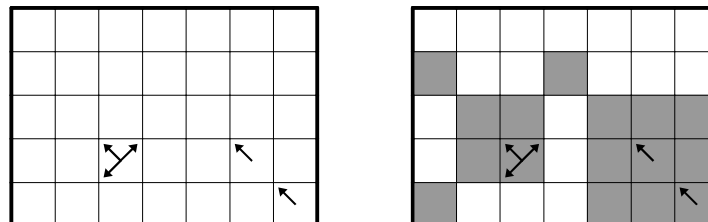
Divide each outlined region into exactly two connected parts: one rectangular and the other of irregular shape. A circle on an edge must touch exactly one rectangle. Borders can not be drawn over cross on edges.



5, 6. Square Chain

20 + 30 points

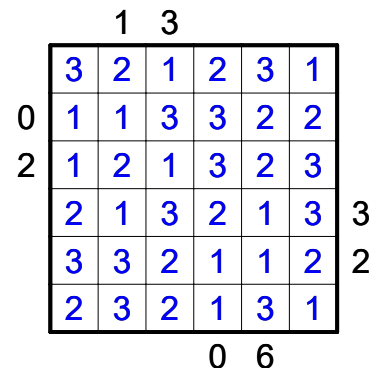
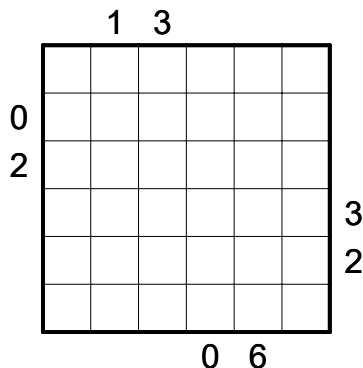
Shade some cells so that every group of orthogonally connected shaded cells must form a filled square. Squares may only ever touch each other at their corners, never along an edge. Certain cells contain a clue in the form of diagonal arrows. A clue cell must be shaded, and its arrows indicate exactly the directions in which the square covering that clue has a diagonally adjacent neighbouring square touching it at a corner. A square may cover no clues or several. Two squares of the same size may not be diagonally adjacent to each other. All shaded cells together must form a single diagonally contiguous area.



7, 8. X-Sandwich

80 + 105 points

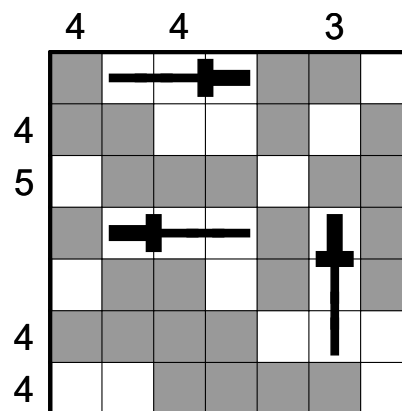
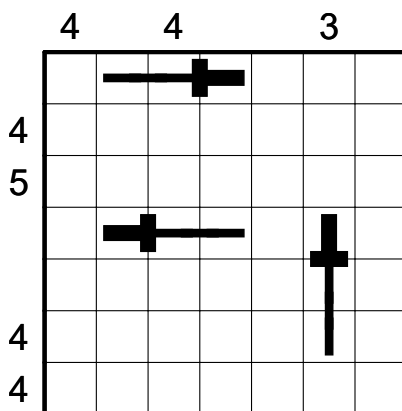
Place a number from 1 to $N/2$ into each cell so that each row and column contains every number from that range exactly twice, where N is the side length of the grid. A clue outside the grid indicates the sum of all numbers between two Xs in the corresponding row or column from the direction of the clue, where X is the first number in that row or column from the direction of the clue.



9, 10. Khanda Thrust

35 + 25 points

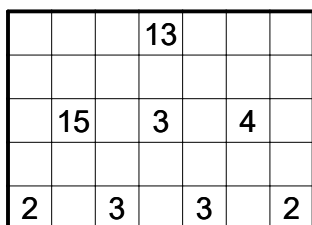
Shade some cells on the board/grid. There is a block of one or more cells at both ends of the sword. The number of cells in the block pointed to by long pointer is twice the number of cells in the block pointed by short pointer. Every block is pointed at by only one sword. Numbers outside the grid, if given, indicate the number of shaded cells.



11, 12. Physarum

30 + 40 points

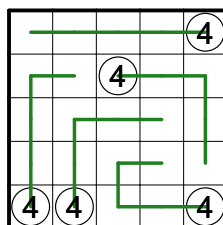
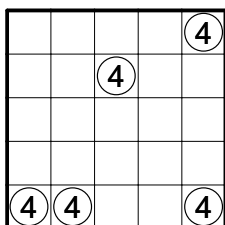
Shade some cells so that shaded cells form orthogonally connected blocks. All shaded blocks must together form a single diagonally connected area, with no isolated block. Every numbered cell must be shaded, and its number equals the count of shaded cells in its eight-cell neighbourhood plus the size of its own shaded block.



13, 14. Urumi Spirals

30 + 50 points

Draw an orthogonal line moving out from each circle that can only travel straight or turn right (i.e. no left turns). Line segments between each corner can only reduce in length or stand the same (i.e. they can never be longer than the previous segment). When finished, every cell will be entered by exactly one line (i.e. the lines cannot intersect themselves or each other). A number in a circle gives the total length of the joined line (i.e. the number of other cells the line visits).



During their stay in the forest Ravana's sister Shoorpanakha approached Rama and Laxman infatuated by their stature. Infuriated by their dismissals, the humiliated and envious Shoorpanakha attacked Sita. Laxman quickly intervened to save Sita by cutting off her nose and ears.

Reflecting the act, this round presents eight puzzles that must be split into two parts, with each part solving as the same puzzle type.

1. Double Crossword

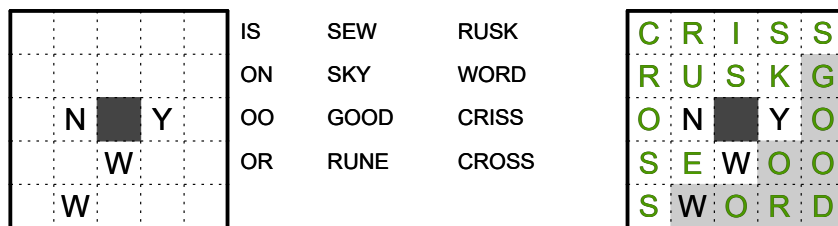
160 points

Crossword Rule:

Fill each of the white cells in the grid going across and down as in crosswords with the given words. All instances of the letters, V and Y (N, W and Y in the example) are already provided in the grid.

Double Rule:

The grid contains two separate puzzles of this type. Each puzzle touches the edge of the grid, and neither puzzle is completely enclosed within the other.



2. Double Yin Yang

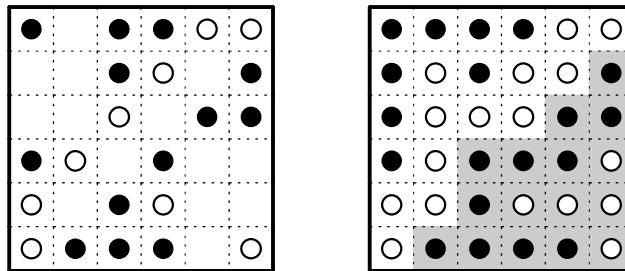
120 points

Yin Yang Rule:

Place a circle into each cell of the grid - some white and some black - such that all circles of the same type must lie in cells forming one orthogonally connected area. No 2x2 region may contain all one type of circle.

Double Rule:

The grid contains two separate puzzles of this type. Each puzzle touches the edge of the grid, and neither puzzle is completely enclosed within the other.



3. Double Statue Park

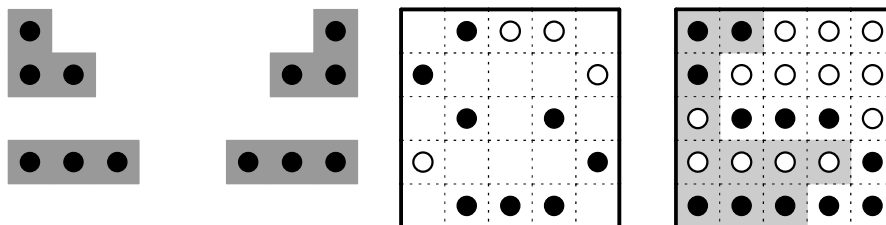
175 points

Statue Park Rule:

Place each shape from the bank given outside the grid into the grid so that no two shapes share an edge and all unused cells form one orthogonally connected area. Rotating and reflecting shapes is allowed. Cells with black circles must be used by a shape, and cells with white circles must not be used by a shape.

Double Rule:

The grid contains two separate puzzles of this type. Each puzzle touches the edge of the grid, and neither puzzle is completely enclosed within the other.



4. Double Rassi Silai

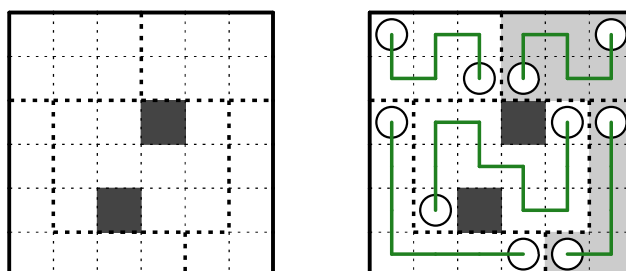
115 points

Rassi Silai Rule:

Within each region, draw a non-intersecting path through the centers of all cells. Paths may not cross over bold borders. No two cells in the grid containing endpoints of paths may be touching one another, not even diagonally. Black cells are not part of any region.

Double Rule:

The grid contains two separate puzzles of this type. Each puzzle touches the edge of the grid, and neither puzzle is completely enclosed within the other.



5. Double Country Road

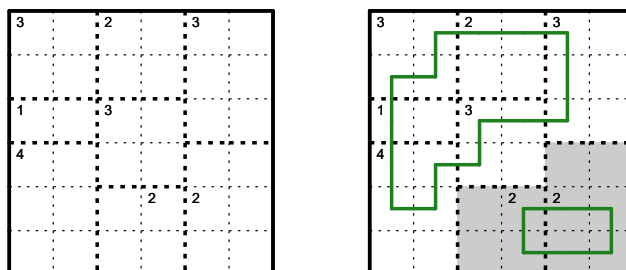
160 points

Country Road Rule:

Draw a non-intersecting loop through the centers of some cells which passes through each region exactly once. A number in a region represents how many cells in the region are visited by the loop. Orthogonally adjacent cells across a region border may not both be unused.

Double Rule:

The grid contains two separate puzzles of this type. Each puzzle touches the edge of the grid, and neither puzzle is completely enclosed within the other.



6. Double Ripple Effect

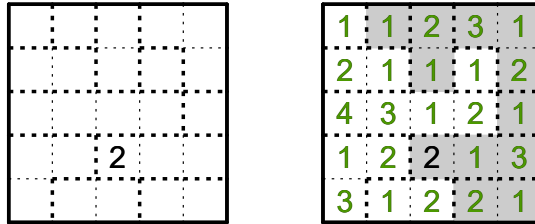
70 points

Ripple Effect Rule:

Place a number into each cell so that each region contains the numbers from 1 to N with no repeats, where N is the number of cells in the region. Two instances of the same number in the same row or column must have at least as many cells between them as the value of the number.

Double Rule:

The grid contains two separate puzzles of this type. Each puzzle touches the edge of the grid, and neither puzzle is completely enclosed within the other.



7. Double Fillomino

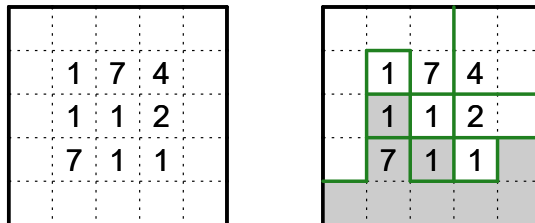
200 points

Fillomino Rule:

Divide the grid into regions of orthogonally connected cells. Two regions of the same size may not share an edge. Clued cells must belong to a region containing the indicated number of cells.

Double Rule:

The grid contains two separate puzzles of this type. Each puzzle touches the edge of the grid, and neither puzzle is completely enclosed within the other.



A humiliated Shoorpanakha fled to the forest demon colony of Janasthana to seek her brother Khara. In response, Khara dispatched an army of fourteen thousand demons to eliminate Rama and Laxman, but Rama single-handedly vanquished them all.

Mirroring this legend, this chapter is structured as a Sprint round comprising 48 puzzles, featuring six easy puzzles across eight distinct puzzle categories.

1, 2, 3, 4, 5, 6. Hitori

5 + 15 + 5 + 10 + 5 + 5 points

Shade some cells so that unshaded do not have repeated symbols in any row or column. Shaded cells do not share an edge and unshaded cells form one orthogonally connected area.

1	1			4	1
3	5	5		3	2
		4		6	5
9	8		9		
4	8		7	7	8
4	5			3	3

1	1			4	1
3	5	5		3	2
		4		6	5
9	8		9		
4	8		7	7	8
4	5			3	3

7, 8, 9, 10, 11, 12. Square Jam

5 + 5 + 5 + 5 + 5 + 5 points

Divide the grid into square regions of orthogonally connected cells. A number indicates the side length of the square it's in. Region borders may not form any four-way intersections.

	1			
				1
	2			
				3

	1			
				1
	2			
				3

13, 14, 15, 16, 17, 18. Minesweeper

5 + 5 + 5 + 10 + 10 + 15 points

Place mines into some empty cells so that each clue has the indicated number of mines in the (up to) eight surrounding cells.

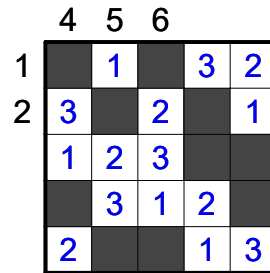
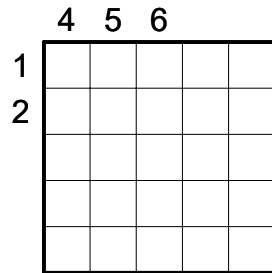
2		1		3	2
2		1		4	1
1		2		4	1

2	☛	1		3	☛	2
☛				☛	☛	
2		1		4		1
☛			☛	☛		
1		2	☛	4	☛	1

19, 20, 21, 22, 23, 24. Doppelblock

10 + 10 + 15 + 15 + 15 + 15 points

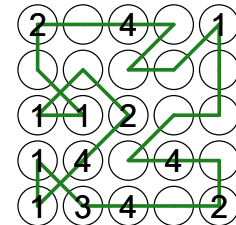
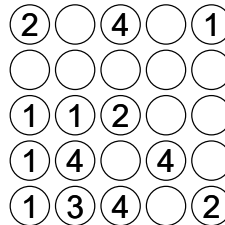
Place a number from 1 to $N - 2$ into some cells so that each row and column contains every number from that range with no repeats, where N is the side length of the grid, and shade the remaining two cells of each row and column. A clue outside the grid indicates the sum of the digits which appear between the two shaded cells in the corresponding row or column.



25, 26, 27, 28, 29, 30. Achtelwenden

10 + 10 + 15 + 15 + 10 + 25 points

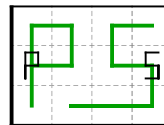
Draw a loop by connecting the centers of orthogonally or diagonally adjacent cells. The loop must travel through every cell once. Diagonal line segments are allowed to intersect. A number in a cell, when multiplied by 45° , indicates the angle formed by the loop in that cell.



31, 32, 33, 34, 35, 36. Curve Data

5 + 5 + 10 + 15 + 15 + 20 points

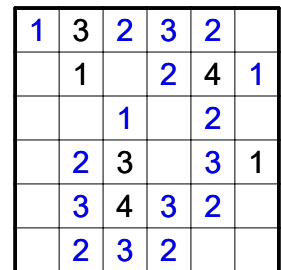
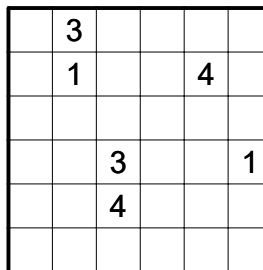
Draw lines between the centers of cells so that each connected figure goes through exactly one clue, and all cells are used by a figure. Clues show how their figures turn and connect with themselves, not allowing rotation or reflection, but do not indicate the lengths of the line segments.



37, 38, 39, 40, 41, 42. Sukoro

10 + 10 + 10 + 15 + 10 + 30 points

Place a number into some cells so that all cells with numbers form one orthogonally connected area. Numbers represent how many of the (up to) four orthogonally adjacent cells contain numbers. No two orthogonally adjacent cells may contain the same number.



43, 44, 45, 46, 47, 48. Boggle

15 + 20 + 30 + 40 + 25 + 25 points

Place a letter in each empty cell such that all the given words can be found in the grid by travelling horizontally, vertically or diagonally. The same cell can be used multiple times within the same word. If a word has the same letter in consecutive positions then that letter has to be placed in two separate cells.

ONE
TWO
THREES



Rama went chasing a golden deer (which is actually the demon Maricha in disguise) and does not return for a long time. At Sita's behest, Laxman set out to search for Rama. However, before leaving he used his divine powers and drew a protective boundary (Laxman Rekha) around the cottage and instructed Sita not to cross it. Following Laxman's departure, Ravana arrived disguised as a mendicant, tricked Sita into crossing the line to offer alms, and abducted her to Lanka. This round features a single loop puzzle with different constraints that change their nature depending on whether they are inside or outside the loop.

Loop Mashup

Draw a single, non-intersecting loop that consists of horizontal and vertical segments along the edges of cells. All clues have a different behaviour based on whether they are inside the loop or outside it. The rules for each type, in terms of these differing behaviours, are mentioned below. Clues inside the loop follow classic rules, while those outside follow a contrasting behaviour.

Myopia : The arrow clues, if inside the loop, indicate all the directions (up, down, left, and right) where the nearest loop segments are located when looking from that cell. However, if the clue is outside the loop, the arrows indicate the locations of the farthest loop segments instead. All arrows must point toward at least one segment.

Line of Sight: The clues with numbered arrows, if inside the loop, indicate the length of the loop segment first seen in the direction of the arrow. However, if the clue is outside the loop, the clue becomes a Knapp Daneben clue with the actual number being +/-1 of the given number.

Polygraph : The numbered clues, if inside the loop, indicate the number of the cell's edges used by the loop. However, if the clue is outside the loop, the clue indicates the number of the cell's edges not used by the loop.

Parallel Counts: Squares placed on an edge between two cells indicate whether the number of unused edges till the next loop segment in that direction (or edge of the grid if no segment) on both sides parallel to that edge are balanced or not. If the square is inside the loop, the counts of unused edges on both sides are equal (balanced). If the square is outside the loop, the counts of unused edges on both sides are unequal (unbalanced). A number inside a square indicates the total number of such unused edges. The loop cannot pass through the squares.

Sheep and Wolf : Sheep marked with S are inside the loop and Wolves marked with W are outside the loop.

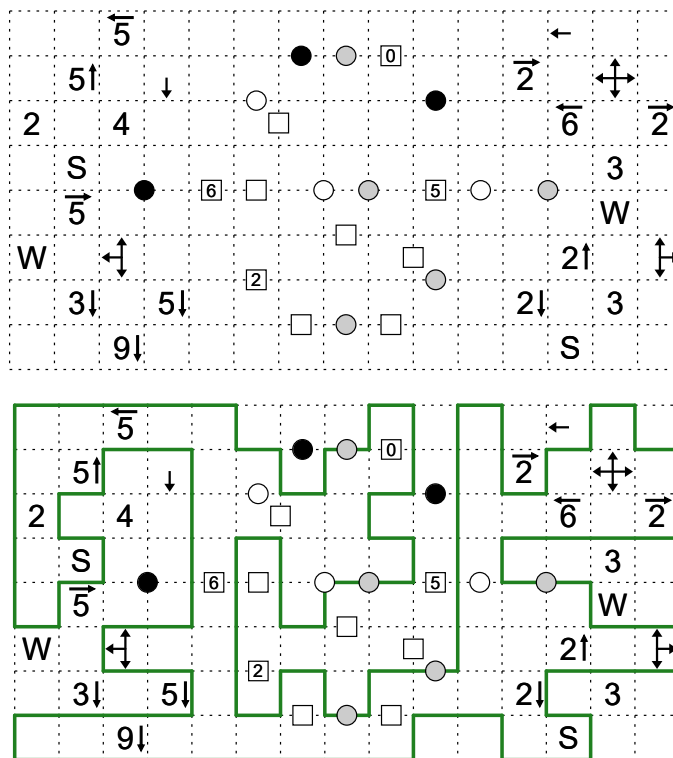
Kurarin: A black dot must touch more cells outside the loop than inside. A white dot must touch more cells inside the loop than outside. A grey dot must touch an equal amount of each. Dots on cell edges touch 2 cells. Dots on cell corners touch 4 cells. The loop can pass through these dots.

	Clue	Inside	Outside
Myopia		Nearest	Farthest
Line of Sight		Correct length	Off by one
Polygraph	1	Edges Used	Edges Unused
Parallel Counts		Balanced	Unbalanced
Sheep/Wolf	S / W	Sheep	Wolf
Kurarin		More cells inside	
		Equal cells inside and outside	
		More cells outside	

The rule summary table above will be available in puzzle booklet.

The competition puzzle grid size is 25x25 cells. If the entire puzzle is not completely correct, while checking, the grid will be divided into 25 areas of 5x5 cells. 15 points will be given for each correctly solved 5X5 area of the puzzle. The partial solution must be part of overall solution.

400 points



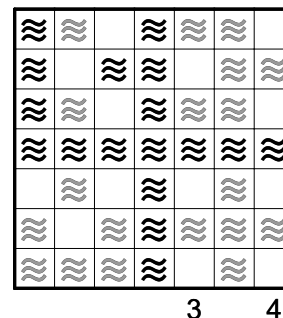
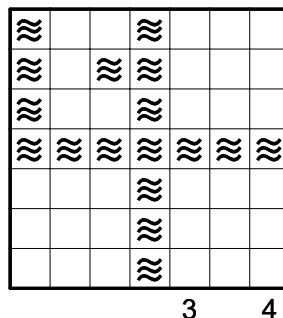
During the search for Sita, Rama and Laxman arrived at Shabri's cottage, where she had been eagerly anticipating their visit for days. Driven by deep devotion and care, she tasted each berry first to ensure only the sweetest and ripest ones were given to Rama.

This round is a curated anthology of creative concepts handpicked by the author team from the past eight WPCs, starting from the previous WPC in India in 2017.

1. Maximal Archipelago [2017]

120 points

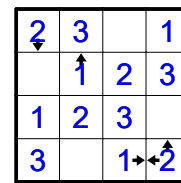
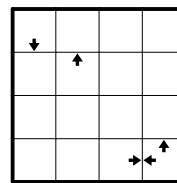
Turn some land squares into sea such that, a) all sea squares are connected to each other through other sea squares orthogonally, b) no two land squares are adjacent orthogonally and c) it is impossible to turn a sea square to land while still satisfying the above two conditions. A number outside the grid indicates the number of land squares remaining in the corresponding row/column.



2. Point to Numbers [2018]

55 points

Place numbers 1-5 (1-3 in the example) in the grid so that each number appears exactly once in each row and column. Some cells will remain empty. If there is an arrow pointing from the cell, it means that the number of digits in this direction corresponds to the number in the cell. Not all possible arrows are given.



3. Permaculture (Shingoki (Full)+ Balance Loop) [2019]

40 points

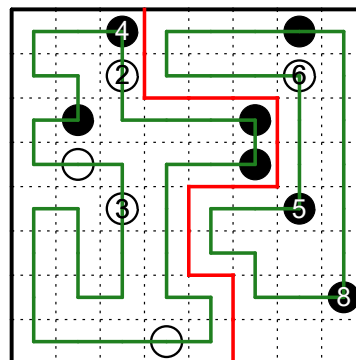
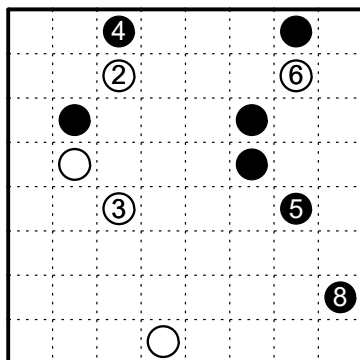
Divide the grid into two parts, so that one part is solved as Shingoki (Full) and the other part as Balance Loop. Within each part all cells must be connected, and each cell must belong to exactly one part. It is possible that one part completely surrounds the other. Aside from this division, the parts are completely independent and do not interact with each other. It is not necessary to draw the borders dividing the two grids.

Shingoki (Full)

Draw a non-intersecting loop through the centers of all cells that passes through every circle. The loop must turn on black circles. The loop must go straight through white circles. Numbers indicate the sum of the segment lengths on both sides of the circle.

Balance Loop

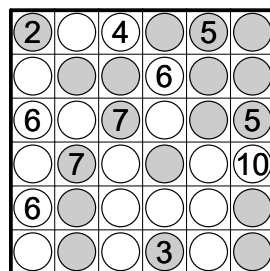
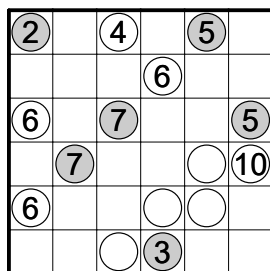
Draw a non-intersecting loop through the centers of some cells that passes through every circle. All white circles must have equal segment lengths on both sides of the circle before turning. All black circles must have unequal segment lengths on both sides of the circle before turning. Numbers indicate the sum of the segment lengths on both sides of the circle.



4. Go [2021]

40 points

Shade some of the empty cells (by placing black go stones). Cells of the same colour that touch each other side by side (horizontally or vertically) form a region. A number shown on a region indicates how many opposite-coloured go stones are orthogonally adjacent to the entire region. Existing go stones cannot change colour.

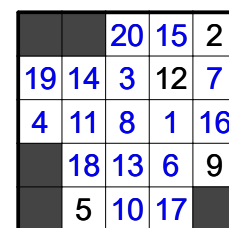
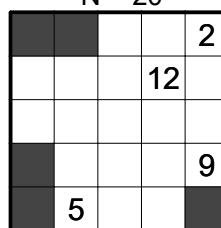


5. Knight Hidato [2022]

25 points

Place a number from 1 to N into each cell so that every number appears once. Every number must be a knight's move away from all numbers in the grid that are consecutive with it.

N = 20



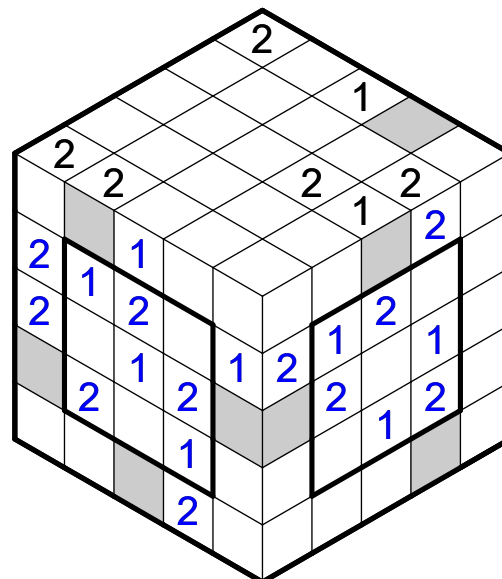
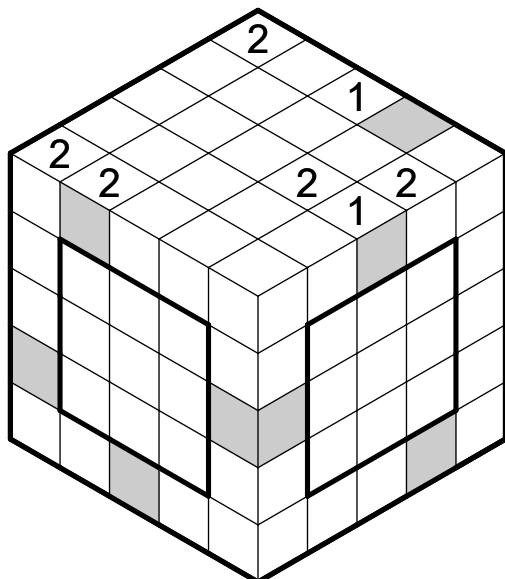
6. Niagara Falls – Gapped Skyscraper [2023]

75 points

Every number at the top of the waterfall must drop onto either its left or right side and land at any open cell outside the central grids ie 4x4 box (3x3 box in example). Numbers cannot pass through or be placed on the stones (shaded cells). The falling numbers cannot be placed in the cells inside the 4x4 box on both sides. These numbers do not block each other and must serve as outside clues for the left and right central grids.

Only one number can be placed in any cell. Two or more numbers can fall in the same direction provided there are enough open cells in that direction.

A Gapped Skyscraper puzzle is created in both left and right grids as each number on the waterfall drops and land in the open cells as the outside clues. Fill every row and column with numbers 1-3 (1-2 in the example) exactly once, leaving one cell blank. Numbers inside represent building heights, while outer clues show visible buildings from that direction. Taller buildings block shorter ones.



7. Quadruple Happiness [2024]

55 points

Shade some cells and draw a single closed loop (without intersections or crossings) through all remaining cells. Shaded cells cannot share an edge with each other. The given clues cannot be part of the loop. Each of the following four types of clues must be satisfied.

Yajilin: Gray cells with arrow

Numbered arrows in such cells indicate the number of shaded cells in given direction.

Castle Wall: Black/White cells with arrow

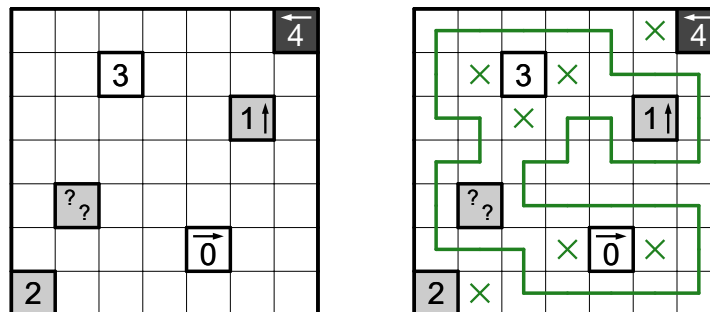
Black cells must be outside the loop. White cells must be inside the loop. Numbered arrows indicate the total sum of the lengths of loop segments in the given direction.

Tapa Like Loop : Gray cells without arrow

Clues inside such cells indicate the number of neighboring cells visited by the loop. If there is more than one number in a cell, each number should be represented with a separate loop segment. A question mark “?” as a clue can represent any positive number.

Koburin : White cells without arrow

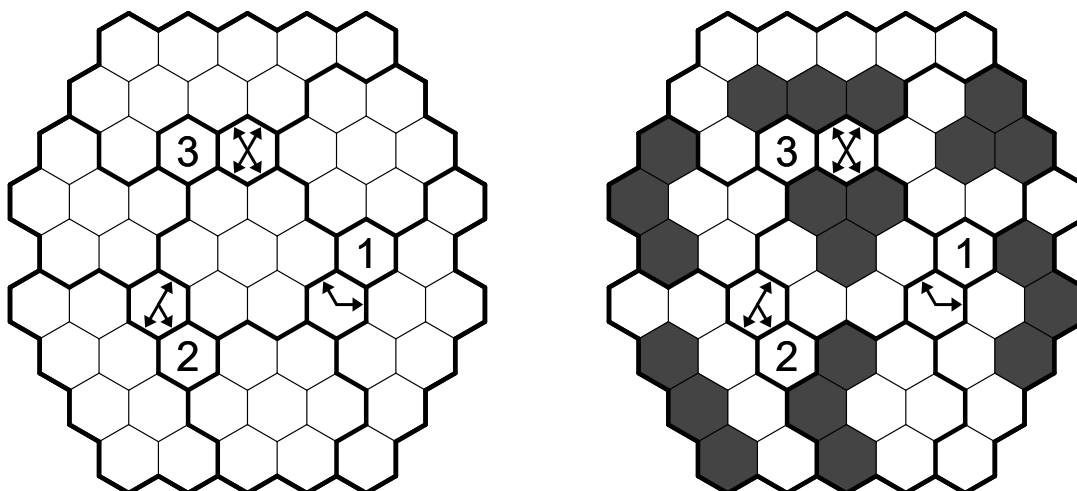
Numbers indicate the count of orthogonally adjacent shaded cells.



8. Hexa Hungary [2025]

40 points

Shade some cells in the grid so that there are exactly six connected shaded cells in every region (three in the example puzzle below). If two adjacent cells belong to different regions, they cannot both be shaded. Number clues indicate the number of the adjacent cells (out of up to six) that are shaded. Arrow clues indicate all directions (out of up to six) where shaded cells appear closest. The arrows' range of visibility is not blocked by clue cells, region or puzzle boundaries. Cells with number clues or arrow clues are considered (and must remain) unshaded, even though they are highlighted. The colouring of the thick lines only serves aesthetic purposes and can be ignored – all thick lines define regions no matter their colour.



Led by Vali's son Angada and Sugriva's minister Hanuman, the search party journeyed toward the South. Reaching the edge of the ocean with no sign of Sita, they were filled with despair. Hanuman however, leaped across the ocean, overcoming numerous obstacles along the way before making his way to Lanka. Following various encounters with the demons there, he successfully delivered Rama's message to Sita and fulfilled his mission. This round is in a relay format with six puzzles from different categories, where clues are shared between consecutive puzzles.

1. Slitherlink

25 points

Connect some pairs of orthogonally adjacent dots to form a single non-intersecting loop. Clues represent the number of edges drawn surrounding the clue (up to four).

A shaded cell is a valid Slitherlink clue and is also a valid Five Cell clue with the same value in the cage at the corresponding place.

2. Five Cells

50 points

A cell with a cage is a valid Five Cells clue and is also a valid Slitherlink clue with the same value in the shaded cell at the corresponding place.

Divide the grid into regions of exactly five cells. The number inside a cell represents how many of its four sides are segments of region borders (including the border of the grid).

This grid and Star Battle have exactly same regions.

3. Star Battle

50 points

This grid and Five Cells have exactly same regions.

Place stars into some cells such that outlined region contains exactly 1 star. Each row and column contain exactly 2 stars (1 star in the example). Stars cannot touch one another, not even diagonally. Some edges are already marked.

The positions of the stars in this grid are same as the positions of the trees in the Tents puzzle.

4. Tents

50 points

The positions of the trees in this grid are same as the positions of the stars in the Star Battle puzzle.

For each tree in the grid, place a tent in an empty orthogonally adjacent cell, connecting to it. Tents may not touch one another, not even diagonally. A clue given outside the grid represents the number of tents in the corresponding row or column.

The positions of the tents in this grid are same as the positions of the circles in the Golem Grad puzzle.

5. Golem Grad

50 points

The positions of the circles in this grid are same as the positions of the tents in the Tents puzzle.

Draw paths through the centers of cells connecting each circle to exactly one other. Paths may not cross each other, themselves, or through cells containing numbers. All cells used by paths (including the cells with circles) must form one orthogonally connected area. Each number represents the size of the area of cells not used by paths that the number is in. No 2x2 region may be entirely used by paths.

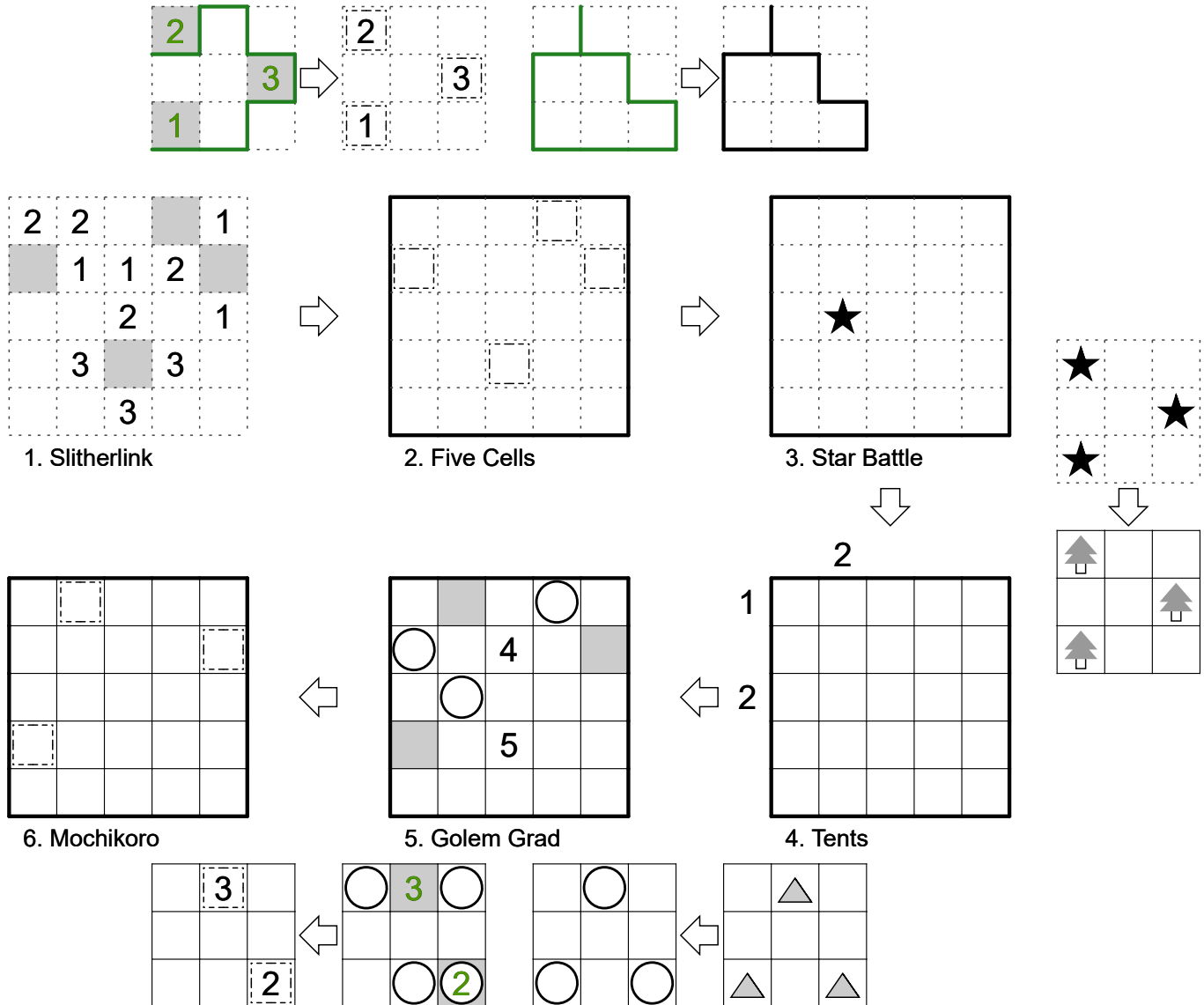
The length of the path passing through the shaded cell in this grid is a valid Mochikoro clue in the cell with the cage at the corresponding place.

A cell with a cage is a valid Mochikoro clue and also indicates the length of the path passing through the shaded cell at the corresponding place in Golem Grad.

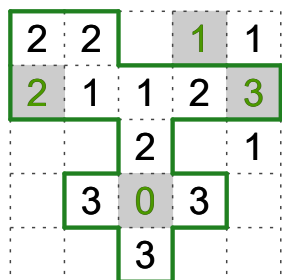
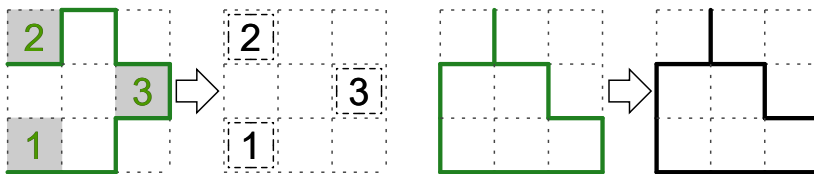
Shade some cells so that all areas of orthogonally connected unshaded cells are rectangular. The unshaded rectangles must all be connected diagonally. Clues cannot be shaded, and represent the number of cells in the unshaded area they belong to. An unshaded area of cells cannot contain more than one clue. No 2x2 region may be entirely shaded.

Example with all types in the relay

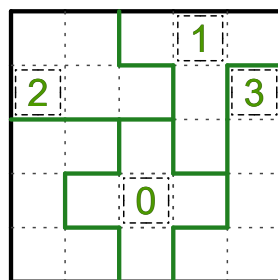
Here is an example with all the six puzzle types in the relay. The small 3x3 grids are not part of the puzzle, they illustrate the clue transfers between grids. The competition puzzle will have identical layout.



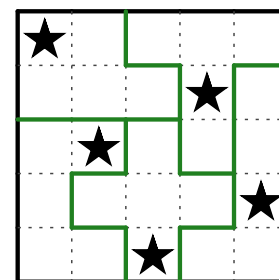
Solutions on next page



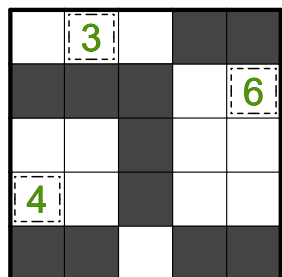
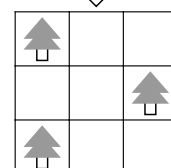
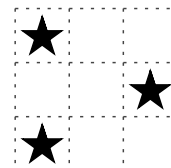
1. Slitherlink



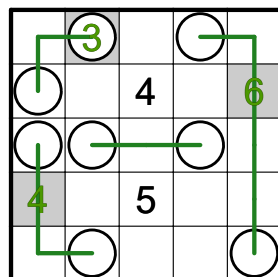
2. Five Cells



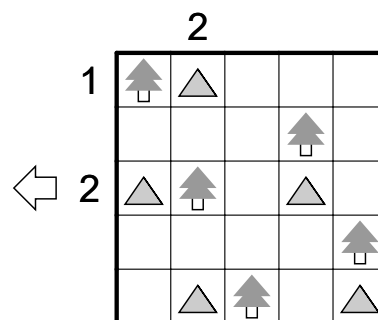
3. Star Battle



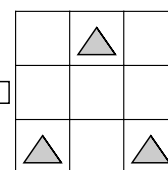
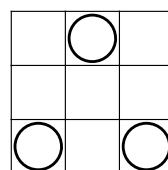
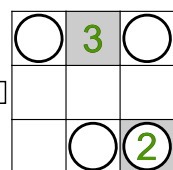
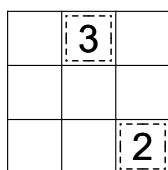
6. Mochikoro



5. Golem Grad



4. Tents



Following Shabri's advice, Rama and Laxman journeyed to Kishkinda, the monkey kingdom's capital, and allied with Sugriva, who was persecuted by his evil twin brother, King Vali. Rama promised to defeat Vali, and Sugriva in exchange pledged to help find Sita. During their duel, Rama hid behind a tree and shot Vali, distinguishing them only because Sugriva wore a garland. Echoing this legend, this round features eight pairs of visually identical puzzles. However, only one puzzle in each pair possesses a unique solution, while the other becomes solvable only after being adorned with a garland.

Round Rule

This round consists of 8 pairs of puzzles. Within each pair, one puzzle has a unique solution, while the other is non-unique with multiple solutions. The non-unique puzzle yields a unique solution only when the given Garland shape is placed into the grid. The Garland cannot overlap any given numbers, letters, dots (even partially), sea cells and arrows. The Garland cannot be rotated or reflected.

Partial points (given within brackets) will be awarded if only one of the two puzzles in the pair is solved correctly and the solution is part of the overall solution.

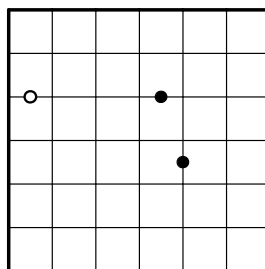
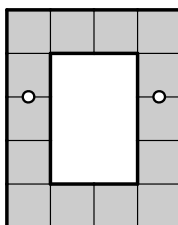
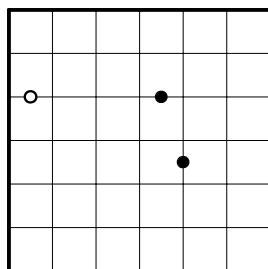
This instruction booklet provides 2 example pairs with Garlands included, while the remaining puzzles appear in their native form.

1. Kropki Loop + Kropki Scrabble

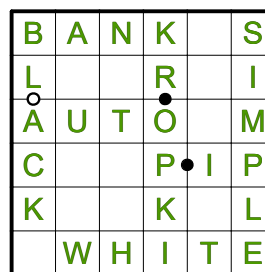
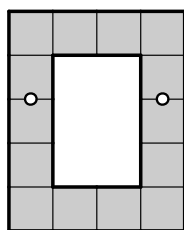
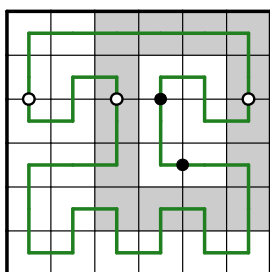
60 (20) points

Kropki Loop

Draw a single closed loop that passes vertically and horizontally through all cells. The loop must go through all of the given dots. When passing through a white dot, the number of cells on one side of the dot before turning must differ by 1 from the number of cells on the other side before turning. When passing through a black dot, the number of cells on one side of the dot before turning must be twice the number of cells on the other side before turning. (Not all possible dots are given; if a line passes through two dots, ignore the other dot and count cells as if it is not there.)



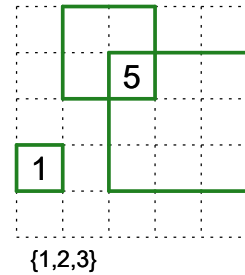
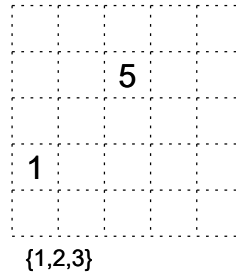
AUTO KROPKI
BANK SIMPLE
BLACK WHITE
PIP



85 (30) points

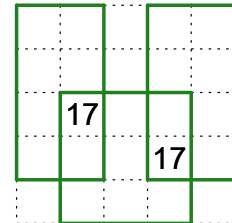
Overlapping Squares

Draw squares with side lengths in the specified list along the dotted grid lines. Each side length must be used exactly as many times as it appears in the list. Squares cannot share any part of their edges or corners, but may intersect each other. Each number given in a cell indicates the sum of the side lengths of the squares that contain that cell. It is possible for a square to contain multiple numbers or no numbers at all.



Overlapping Rectangles

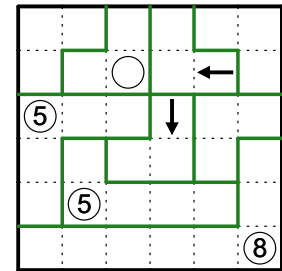
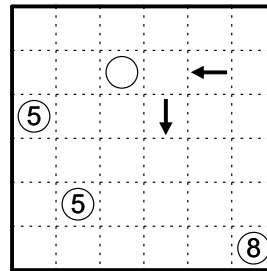
Draw rectangles with each side being at least 2 units along the dotted gridlines. Rectangles cannot share any part of their edges or corners, but may intersect each other. Each number given in a cell indicates the sum of the areas of the rectangles that contain that cell.



5. Sashigane + Sashikabe

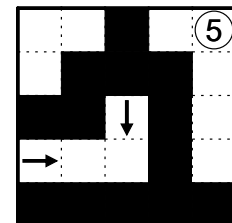
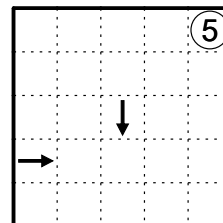
Sashigane

Divide the grid into L-shaped blocks, each one cell wide. Every cell belongs to exactly one block. A circle marks the cell at the corner of its block, and a number in a circle indicates the number of cells in that block. An arrow marks a cell at one end of its block and points towards the corner.



Sashikabe

Place one cell wide L-shaped blocks in the grid so that they do not touch each other orthogonally. A circle marks the cell at the corner of its block, and a number in a circle indicates the number of cells in that block. An arrow marks a cell at one end of its block and points towards the corner. Shade the remaining grid cells to form a single orthogonally connected area. Clues cannot be shaded and no 2x2 region may be entirely shaded.

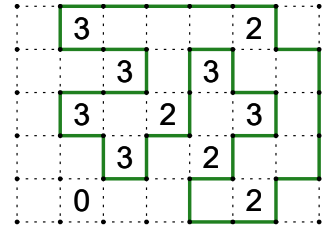
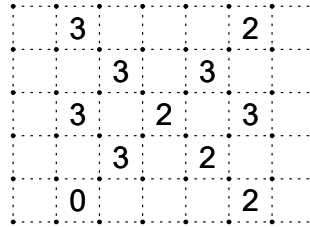


6. Slitherlink + Vertex Slitherlink

50 (15) points

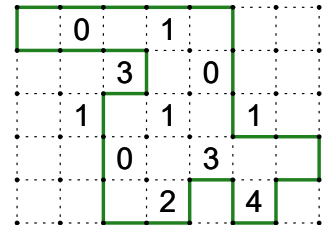
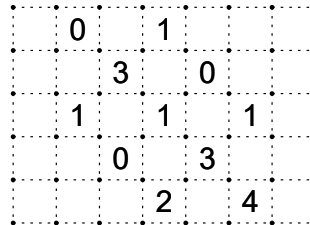
Slitherlink

Connect some pairs of orthogonally adjacent dots to form a single non-intersecting loop. Clues represent the number of edges drawn surrounding the clue (up to four).



Vertex Slitherlink

Connect some pairs of orthogonally adjacent dots to form a single non-intersecting loop. Clues represent the number of turns the loop makes on the four surrounding vertices.

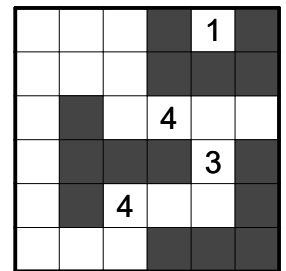
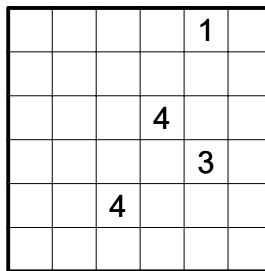


7. Pentasight + Tetrasight

105 (35) points

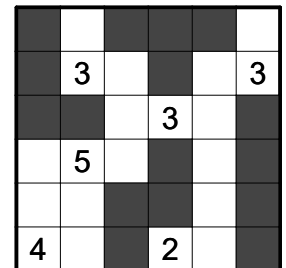
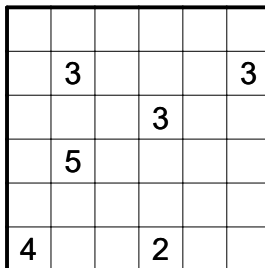
Pentasight

Place unique pentominoes (rotations and reflections count as the same shape) in the grid so they do not touch, even diagonally. Grid numbers show the total horizontally and vertically visible white cells from that cell, including itself, with pentominoes blocking the line of sight.



Tetrasight

Place unique tetrominoes (rotations and reflections count as the same shape) in the grid so they do not touch, even diagonally. Grid numbers show the total horizontally and vertically visible white cells from that cell, including itself, with tetrominoes blocking the line of sight.

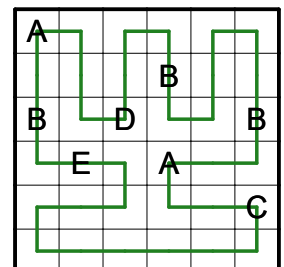
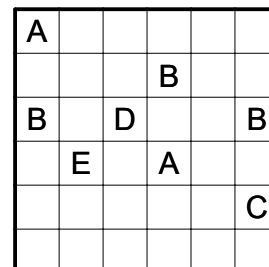


8. Gemini Loop + Gemini Regions

65 (25) points

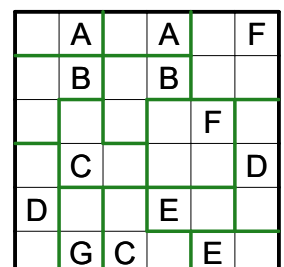
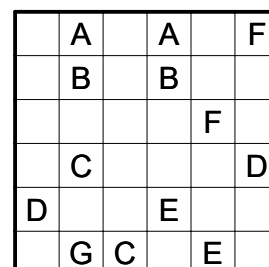
Gemini Loop

Draw a closed loop passing through centers of all cells horizontally or vertically. Cells with the same letter all have the same appearance as to how the loop enters and exits the cell. Cells with different letters have different appearances.



Gemini Regions

Divide the grid into regions with minimum two cells such that each region has exactly one letter. The regions with the same letter have the same shape and the letter is placed in the same position within the region. Regions with different letters either have different shapes or different position of the letter within the shape. Rotations and reflections are not counted as same shape.



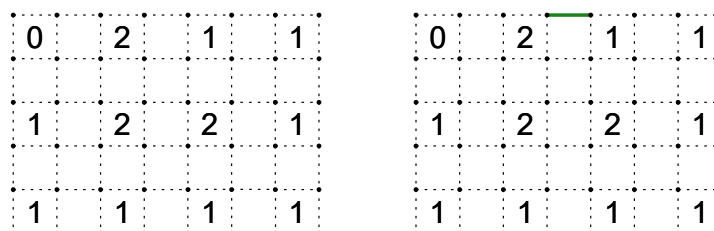
While the mighty monkeys hauled huge boulders and logs to build the bridge, a tiny squirrel, equally determined to help, would wet its body, roll in shore sand, and shake it into the crevices between the giant rocks. Though the monkey army mocked the squirrel, Rama observed that the sand bound the stones, making the bridge stronger. He explained that every effort, however small, contributes to the fulfillment of the overall task and is vital to completing the collective mission. Drawing inspiration from this tale where every effort matters, this round calls for participants to solve just one element for each puzzle from varying categories.

1. Slitherlink

5 points

Connect some pairs of orthogonally adjacent dots to form a single non-intersecting loop. Clues represent the number of edges drawn surrounding the clue (up to four).

Squirrel Rule: There is exactly one loop segment that can be drawn logically.

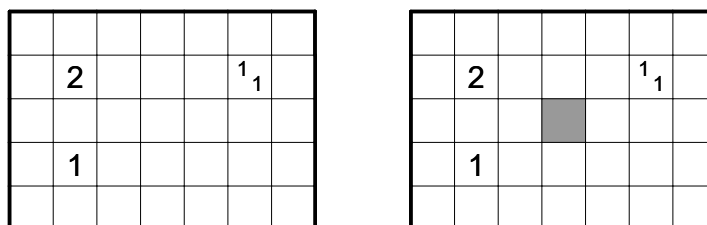


2. Tapa

35 points

Shade some cells so that all shaded cells form one orthogonally connected area. Clues cannot be shaded, and represent the lengths of the blocks of consecutive shaded cells in the (up to) eight cells surrounding the clue. If there is more than one number in a cell, then there must be at least one white (unshaded) cell between the shaded blocks. No 2x2 region may be entirely shaded.

Squirrel Rule: There is exactly one cell that can be shaded logically.

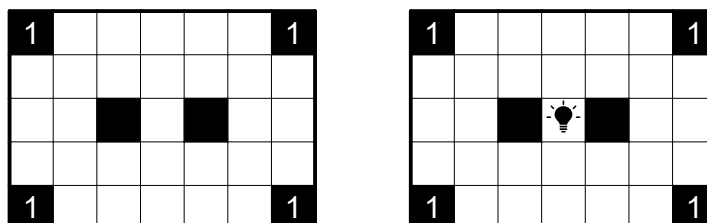


3. Akari

15 points

Place light bulbs in some empty cells so that every cell is illuminated. A light bulb illuminates the cell it is in and all cells seen in a straight line horizontally or vertically, not obstructed by a black cell. No light bulb may illuminate another. Clues represent the number of light bulbs in the (up to) four cells surrounding the clue.

Squirrel Rule: There is exactly one cell in which a light bulb can be placed logically.

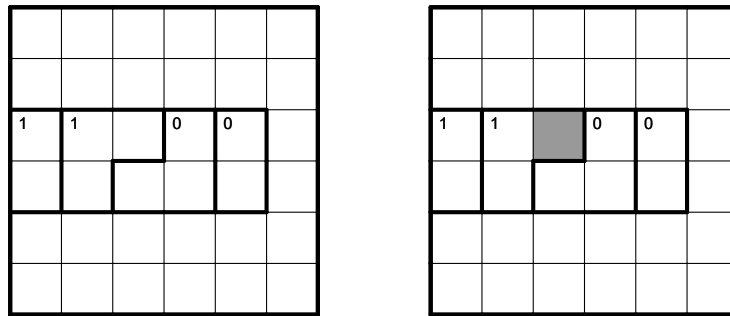


4. Regional Yajilin

10 points

Shade some cells and draw a single closed loop (without intersections or crossings) through all remaining cells. Shaded cells cannot share an edge with each other. A number in a region indicates how many cells must be shaded in that region. Cells with number clues may be shaded.

Squirrel Rule: There is exactly one cell that can be shaded logically.

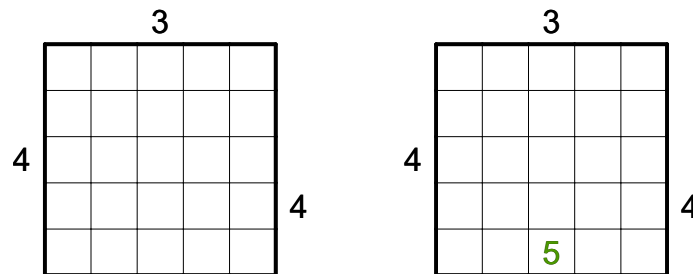


5. Skyscraper

5 points

Place a number from 1 to N into each cell so that each row and each column contains every number from that range with no repeats, where N is the side length of the grid. Each number inside the grid represents the height of the skyscraper in that cell. Each number outside the grid represents the number of skyscrapers that can be seen in the corresponding row or column. Taller skyscrapers hide shorter ones.

Squirrel Rule: There is exactly one cell in which a number can be placed logically.

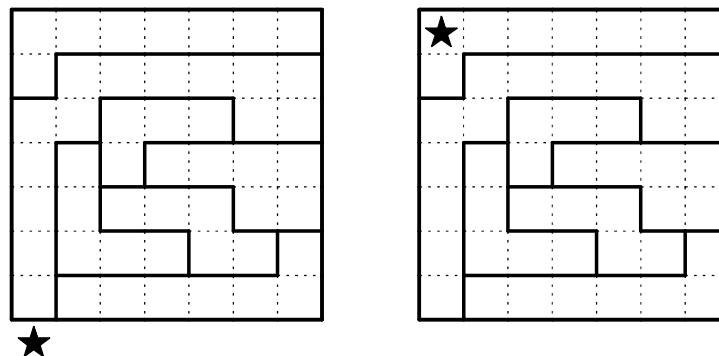


6. Star Battle

20 points

Place stars into some cells such that each row, column, and outlined region contains given number of stars. Stars may not touch one another, not even diagonally.

Squirrel Rule: There is exactly one cell in which a star can be placed logically.

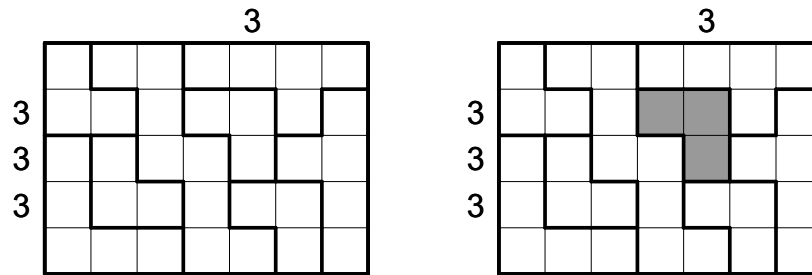


7. Crazy Pavement

35 points

Shade all cells of some regions. Numbers outside the grid indicate the number of shaded cells in that row/column.

Squirrel Rule: There is exactly one region that can be shaded logically.

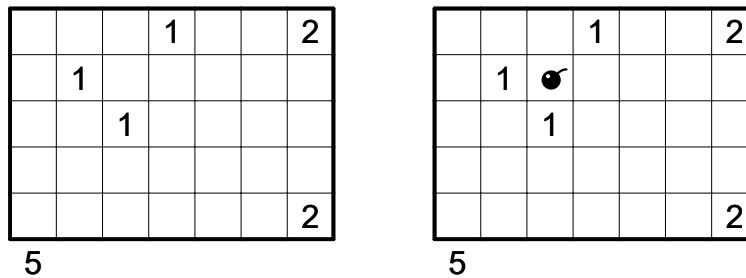


8. Minesweeper

15 points

Place the number of mines given below the grid into some empty cells in the grid such that the numbers in the grid represent the number of mines in the (up to) 8 neighboring cells, including diagonal ones.

Squirrel Rule: There is exactly one cell in which a mine can be placed logically.

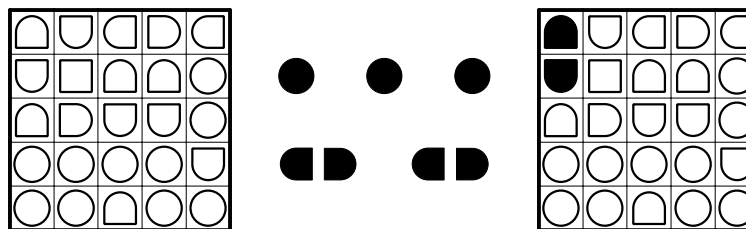


9. Retrograde Battleships

30 points

Place the given fleet of ships with the shapes of the ships as shown. Ships cannot touch each other, even diagonally. Possible outlines of ship segments are given in the grid. If a cell is used, it must be of that given segment.

Squirrel Rule: There is exactly one full ship that can be placed logically.



10. Hitori

15 points

Shade some cells so that unshaded do not have repeated symbols in any row or column. Shaded cells do not share an edge and unshaded cells form one orthogonally connected area.

Squirrel Rule: There is exactly one cell that can be shaded logically.

1	1					
1						
						4
					2	3

1	1					
1						
						4
					2	3

11. Sudoku

15 points

Place a digit from 1 to 9 into each empty cell in the grid so that each digit appears exactly once in each row, column and 3X3 region.

Squirrel Rule: There is exactly one cell in which a digit can be placed logically.

				6				
1		3		2		8		6
	2						4	
		5				9		
	6		1		5		7	
8				4				3

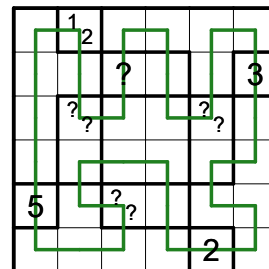
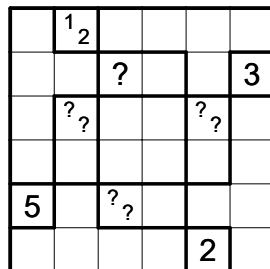
				6				
1		3		2		8		6
	2						4	
		5				9	6	
	6		1		5		7	
8				4				3

As the battle took heavy casualties and Ravana faced imminent defeat, he ordered the awakening of his mighty brother, Kumbhakarna. Kumbhakarna was famous for his immense size, huge appetite, and long periods of sleep. Cursed by Brahma to sleep for six months and wake for only one day, the giant was roused using loud drums, trumpets, and elephants. Kumbhakarna then led a massive army, slaughtering the monkeys until Rama killed him in battle using divine arrows. This Marathon round features seven Kumbhakarna-sized puzzles from different categories.

1. Rail Pool

150 points

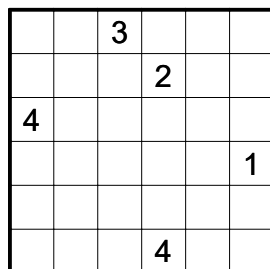
Draw a single, non-intersecting loop that passes through the centers of all cells, including clue cells. Clues represent all of the different lengths of the straight line segments that pass through all cells within the region, either partially or fully contained by that region. Each number within a region must be represented by at least one line segment. Each “?” represents a positive integer; if a clue cell has more than one “?”, all numbers and “?”s in that clue must be different from each other.



2. Nurikabe

140 points

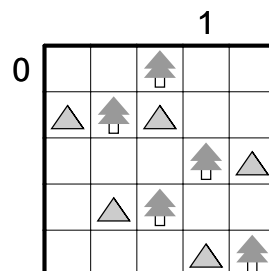
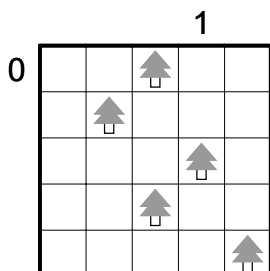
Shade some cells so that all shaded cells form one orthogonally connected area and no 2x2 region is entirely shaded. Clues cannot be shaded, and every orthogonally connected area of unshaded cells contains exactly one clue, the value of which represents the size of the area.



3. Tents

140 points

For each tree in the grid, place a tent in an empty orthogonally adjacent cell, connecting to it. Tents may not touch one another, not even diagonally. A clue given outside the grid represents the number of tents in the corresponding row or column.



4. Five Cells

150 points

Divide the grid into regions of exactly five cells. The number inside a cell represents how many of its four sides are segments of region borders (including the border of the grid).

3		2		2
2		3		3
2				2
3		3		2
3		1		3

3		2			2
2		3			3
2					2
3		3			2
3		1			3

5. Multi Sum Skyscrapers

200 points

Place a digit from 1 to 5 into each empty cell of every 5x5 grid so that no digit repeats in any row or column inside the grid. Each number inside the grid represents the height of a building. The clues on the shaded cells outside the grid indicate sum of the heights of the buildings that can be “seen” when looking from that direction. Taller buildings block the view of smaller buildings.

A shaded cell that lies between two grids is a clue for both and its clue must hold for each of grids.

[The example below uses four 3X3 grids.]

A 6x6 grid with some cells shaded gray and others white. The grid is divided into four 3x3 quadrants by a central 2x2 white square. The top-left and bottom-right quadrants have their top and bottom rows shaded gray. The top-right and bottom-left quadrants have their top and bottom rows white. The central 2x2 square is white. The numbers 5, 4, and 6 are placed in the top-left, top-right, and bottom-left quadrants respectively.

	5	6	3		3	6	5	
5	2	1	3	3	3	1	2	5
3	3	2	1	6	1	2	3	3
4	1	3	2	5	2	3	1	4
	4	3	5		5	3	4	
4	1	3	2	5	2	3	1	4
3	3	2	1	6	1	2	3	3
5	2	1	3	3	3	1	2	5
	5	6	3		3	6	5	

6. Skip Word Search

180 points

Find all the given words in the grid in any of the eight directions such that every consecutive pair of letters in each word has a single skipped cell between it.

A	X	T	W	O	T	ALL
F	A	N	B	O	X	TWO
B	L	A	O	N	E	FIT
I	L	F	L	W	I	IN
N	I	E	X	E	B	ONE
T	F	W	T	O	L	BOX

A	X	T	W	O	T
F	A	N	B	O	X
B	L	A	O	N	E
I	L	F	L	W	I
N	I	E	X	E	B
T	F	W	T	O	L

7. Double Cross

240 points

Place the given pairs of words in the grid. Each pair consists of two words: one written horizontally from left to right, and the other written vertically from top to bottom. These two words intersect at exactly one shared letter, and the locations of these shared letters are marked on the grid. Additionally, letters belonging to different word pairs cannot touch one another, not even diagonally.

UNO--ONE
DOS--TWO
TRES--THREE

T	R	E	S		T	
H					W	
R		U		D	O	S
E		N				
E		O	N	E		

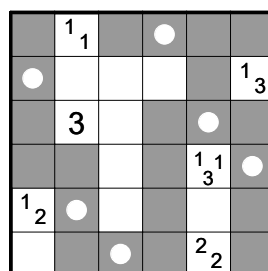
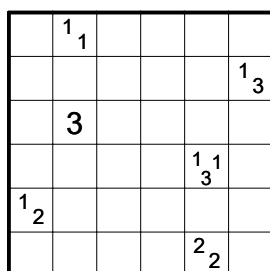
Following Kumbhakarna's defeat, Ravana ordered his son Indrajit, a master of illusion warfare, to re-enter the battle. Vanishing into the clouds, Indrajit rained lethal arrows upon the bewildered monkey army. Vibhishana disclosed that Indrajit attained invincibility through a secret ritual, leaving him vulnerable only while performing it. Armed with this knowledge, Laxman and Vibhishana attacked the temple and, after a fierce battle, Laxman struck Indrajit down using the powerful Indrastra weapon. Mirroring Indrajit's deceptive abilities, his illusory magic has been cast over this round's puzzles, rendering some of the puzzle elements invisible.

1. Invisible Tapa

110 points

Shade some cells so that all shaded cells form one orthogonally connected area. Clues cannot be shaded, and represent the lengths of the blocks of consecutive shaded cells in the (up to) eight cells surrounding the clue. If there is more than one number in a cell, then there must be at least one white (unshaded) cell between the shaded blocks. No 2x2 region may be entirely shaded.

Invisible Rule: There is one shaded cell in each row and column that is invisible to the clues. These shaded cells count for connectivity and the 2x2 restriction.

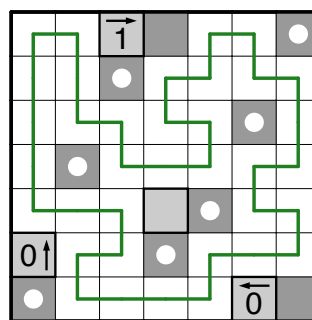
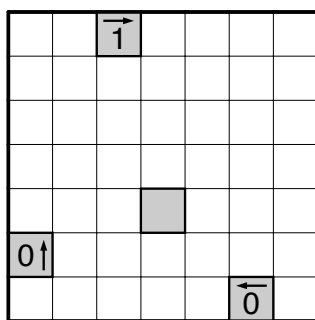


2. Invisible Yajilin

95 points

Shade some cells and draw a single closed loop (without intersections or crossings) through all remaining cells. Shaded cells cannot share an edge with each other. Some cells are outlined and in gray and cannot be part of the loop. Numbered arrows in such cells indicate the number of shaded cells in that direction.

Invisible Rule: There is one shaded cell in each row and column that is invisible to the clues. These shaded cells still cannot share an edge with other shaded cells.

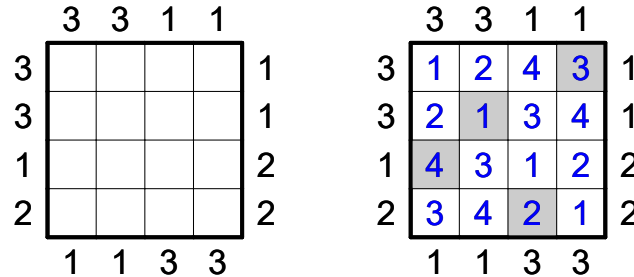


3. Invisible Skyscraper

70 points

Place a number from 1 to N into each cell so that each row and each column contains every number from that range with no repeats, where N is the side length of the grid. Each number inside the grid represents the height of the skyscraper in that cell. Each number outside the grid represents the number of skyscrapers that can be seen in the corresponding row or column. Taller skyscrapers hide shorter ones.

Invisible Rule: Additionally in each row and column there is one invisible skyscraper. There is exactly one invisible skyscraper of each number. The height of each invisible skyscraper is 0.

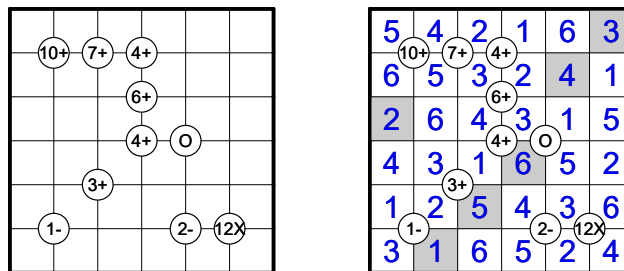


4. Invisible Mathrax

140 points

Place a number from 1 to N into each cell so that each row and each column contains every number from that range with no repeats, where N is the side length of the grid. Clues are given at the intersection of four cells. A number paired with a sign (+, -, x, /) indicates that the result of the operation on both diagonal pairs of cells around the circle equals that number. For subtraction and division, the smaller number is subtracted from or divided from the larger number. Clue "O" or "E" indicates that all four adjacent cells touching the circle must contain odd or even numbers, respectively.

Invisible Rule: There is one invisible digit in each row and column. The invisible digits are all different. If a clue encounters an invisible digit, it is counted as 'blank' so the other digit will be counted as the answer. An invisible digit is can be consider either as odd or as even.

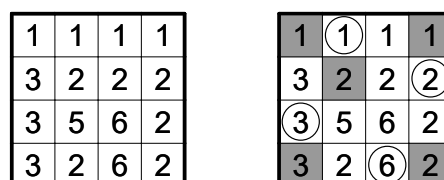


5. Invisible Hitori

115 points

Shade some cells so that unshaded do not have repeated numbers in any row or column. Shaded cells do not share an edge and unshaded cells form one orthogonally connected area.

Invisible Rule: There is one invisible number in each row and column. The invisible numbers are all different. An invisible cell's number is ignored when checking for repeated symbols in unshaded cells. The invisible cells count for connectivity.

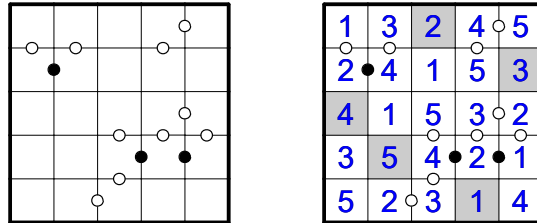


6. Invisible Kropki

35 points

Place a number from 1 to N into each cell so that each row and each column contains every number from that range with no repeats, where N is the side length of the grid. Adjacent cells marked with a white circle contain consecutive digits. Adjacent cells marked with a black circle contains digits where one digit is double of the other digit. The circle between 1 and 2 can be of either colour. All possible circles are marked.

Invisible Rule: There is one invisible digit in each row and column. The invisible digits are all different. Every dot touching an invisible digit has been removed.

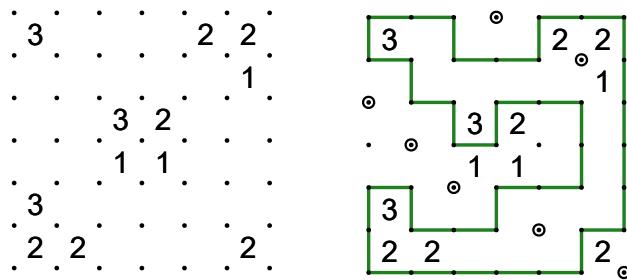


7. Invisible Slitherlink

30 points

Connect some pairs of orthogonally adjacent dots to form a single non-intersecting loop. Clues represent the number of edges drawn surrounding the clue (up to four).

Invisible Rule: In every row and column, there is exactly one dot that is invisible. The loop cannot pass through the invisible dots.

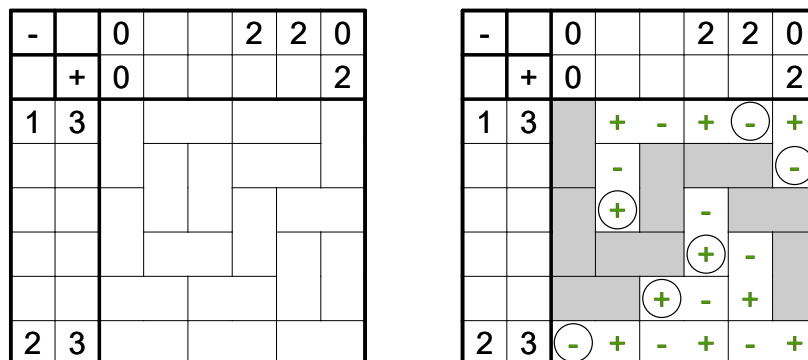


8. Invisible Magnets

55 points

Place a plus sign and a minus sign in some two-cell regions (one sign per cell) of the main grid, so that no two adjacent cells contain the same sign. Number clues outside the grid indicate the number of plus signs or minus signs in the row or column.

Invisible Rule: In each row and column there is a symbol that is invisible to outside clues. The symbol must still be consistent for magnet polarity adjacency rules.



Having witnessed the demise of all his trusted warriors, Ravana prepared for the final battle with Rama and Laxman, setting forth with his generals and entire army to kill Rama. In the intense battle that followed, Rama's arrows shattered Ravana's flag-staff, and Rama finally shot an arrow that severed Ravana's head. To his astonishment, however, a new head immediately appeared in its place, a cycle that repeated with every subsequent strike. Vibhishana disclosed the secret of Ravana's nectar of immortality, which was stored in his belly. Armed with this knowledge, Rama used a powerful celestial arrow from Sage Agastya to strike Ravana's navel, finally killing the ten-headed demon king.

To commemorate this legendary victory, the final playoffs will see eleven contenders race through 10 puzzles, with each puzzle representing one of Ravana's ten heads. One competitor will be eliminated at every puzzle, and the ultimate survivor will be crowned the "Puzzle Playoff Champion".

Playoffs Format

This final round is the individual playoffs which will be in a knock out horse race format between the top 11 participants from the individual rounds. The top 11 seeds will be determined based on the total points in the individual rounds. In case of a tie, the sum of ranks in individual rounds will be used to determine the seeds. In case there is still a tie, a randomizer will be used to determine the seeds.

There will be a total of 10 stages in the playoffs, with one puzzle in each stage. All 11 participants will start the first stage with a time advantage for the higher seeded participants. The top 3 seeds will start the playoffs, with seeds 4 to 7 starting 30 seconds later and seeds 8 to 11 a further 30 seconds later.

After finishing the puzzle participants will submit to the invigilators who will check the puzzle within a checking window of 30 seconds. If the puzzle is incorrect, the sheet will be handed back to the participant after the checking window is over and can be submitted again after correction. If the puzzle is correct the participant will transfer to the next stage and can start solving the puzzle after a transfer window of 30 seconds. The table that participant vacates will be not accessible for rest of the participants.

In each subsequent stage, there will be one less table than the previous stage and hence the last participant to arrive at that stage will be eliminated.

For clearing each stage participants will be awarded bonus points according to the following table

Stages 1-4	50 points each
Stages 5-7	75 points each
Stage 8	100 points
Stage 9	125 points
Stage 10	150 points

The playoffs will be completed either when the playoff duration of 60 minutes is over or once 10 of the 11 participants are eliminated from the race.

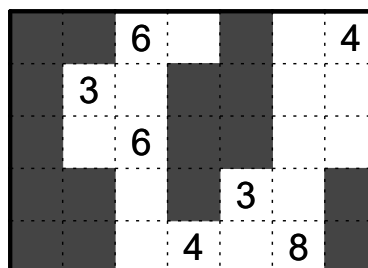
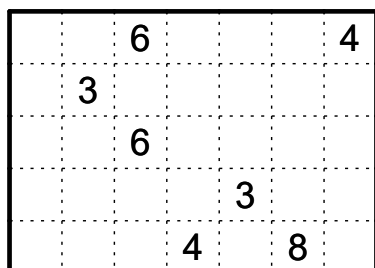
The points earned by the participants across the playoff rounds will be added to their base points from the individual rounds to crown the World Puzzle Champion.

1. Cave X 4

30 points

Shade some cells to leave behind a single connected group, known as the cave. There should not be any shaded cells enclosed within the cave. All numbered cells must be a part of the cave, with each number indicating the total count of cells connected vertically and horizontally to the numbered cell including the cell itself.

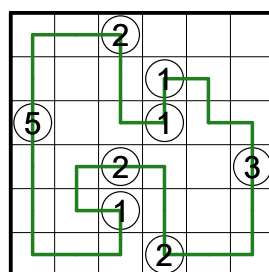
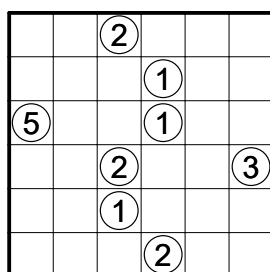
There will be 4 independent Cave puzzles.



2. Geradeweg

30 points

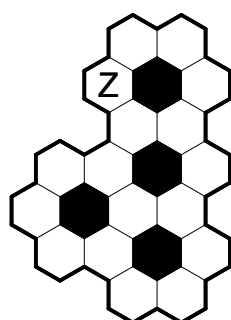
Draw a single, non-intersecting loop that passes through all cells with circles; the loop may either go straight through or turn at each circle. If the loop goes straight through a circle, the number on that circle indicates the length of the straight segment. If the loop turns at a circle, the number on that circle indicates the length of both loop segments extending from that circle.



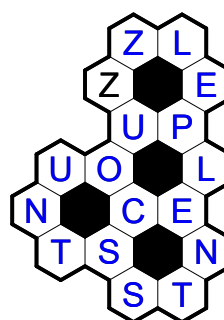
3. Hexa Words

30 points

Fill in the whole grid with letters. The six hexagonal cells around each shaded cell must contain one of the words from the given list. The word may start from any cell and can be read either clockwise or counter-clockwise. Each word should be used exactly once.



COUNTS
COUPLE
PUZZLE
SCENTS



4. TomTom

30 points

Place a number from 1 to N into each cell so that each row and column contains every number from that range with no repeats, where N is the side length of the grid. A clue represents the value obtained by applying an operation iteratively on the numbers in the region the clue is in. If no operation is given, it may be any of +, -, x, or /. Subtraction and division in regions with more than two numbers are handled by taking the largest number and subtracting/dividing all the others.

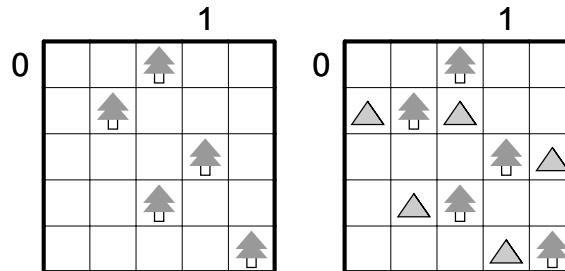
16		8	2	
	8			
			60	
0	1			1

16	1	4	8	3	2	5
	4	8	5	2	3	1
	2	1	4	60	5	3
0	5	1	3	1	4	1
	3	2	5	1	4	

5. Tents

30 points

For each tree in the grid, place a tent in an empty orthogonally adjacent cell, connecting to it. Tents may not touch one another, not even diagonally. A clue given outside the grid represents the number of tents in the corresponding row or column.



6. Knight Hidato

30 points

Place a number from 1 to N into each cell so that every number appears once. Every number must be a knight's move away from all numbers in the grid that are consecutive with it.

N = 20

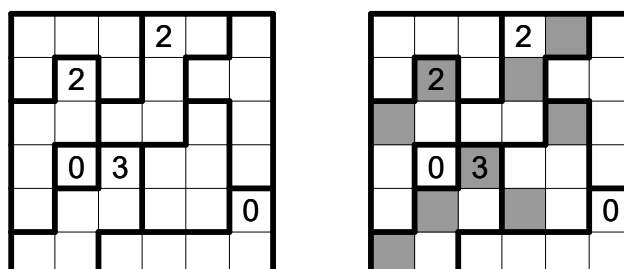
				2
			12	
				9
	5			

		20	15	2
19	14	3	12	7
4	11	8	1	16
	18	13	6	9
	5	10	17	

7. Heyawake

30 points

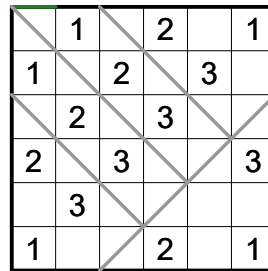
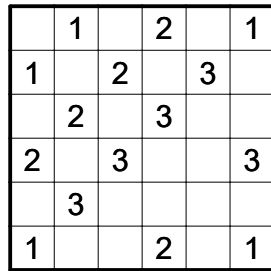
Shade some cells black so that all remaining white cells are connected as part of a single connected group; shaded cells cannot share an edge. No unbroken sequence of white cells in any row or column can cross two thick boundaries. A number in a region indicates the number of shaded cells in that region. Regions with no number may have any number of shaded cells. Cells with numbers can be shaded over.



8. Slash Pack

30 points

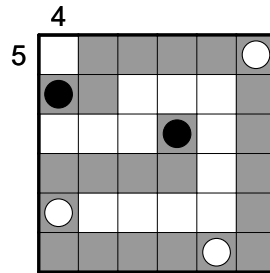
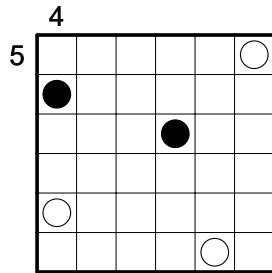
Divide the grid into regions by adding diagonals into empty cells. Two diagonals cannot cross in one cell, and there can be no loose ends. Each region must contain the numbers 1 to N exactly once (where N is the largest number in the grid.)



9. Snake

30 points

Locate a snake (a 1-cell wide path) of unknown length in the grid, whose head and tail are given (Black Circles). The snake does not touch itself, even diagonally. Numbers outside the grid indicate the number of snake cells in that row/column. White circles are part of the snake and must be used.



10. Araf

30 points

Divide the grid into regions of orthogonally connected cells. Each region must contain exactly two circles and have an area that lies strictly between the two numbers in the circles.

